MSUANet: Multistage Information Sharing Multitask Hyperspectral Classification Network With Unmixing Assistance

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***Abstract*—Multitask learning (MTL) can effectively exploit the complementary information across different tasks in hyperspectral images (HSIs) to support classification. However, existing MTL methods for HSI classification usually employ shared structures in the early stages or simple information fusion in the later stages, failing to adequately characterize task-specific feature differences and adaptively select task-relevant information. Therefore, we propose a multistage information sharing multitask hyperspectral classification network with unmixing assistance (MSUANet). The proposed method adopts a three-stage strategy, including low-level information sharing, task-specific information mining, and adaptive latent information selection, to fully exploit the information gain introduced by the unmixing task and provide more fine-grained spectral details for improved classification performance. In addition, we propose an information diverter with hybrid attention (IDHA) module, which enhances task-specific information for classification and unmixing, respectively. Furthermore, we design the adaptive latent abundance knowledge transfer (ALAKT) module, which adaptively selects latent complementary information for both tasks. The experimental results on Indian Pines, Houston 2013, Berlin, and Washington DC MALL datasets demonstrate that MSUANet outperforms state-of-the-art methods. The source code will be available at https://github.com/chihuajiao/HSIMTL**

***Index Terms*—Deep learning (DL), hyperspectral image (HSI) classification, hyperspectral unmixing, multitask learning (MTL).**

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I. Introduction

REMOTE sensing technology [1] has advanced rapidly in recent years, which has significantly enhanced both the spatial and spectral resolution of earth surface observations. Compared with optical remote sensing imagery, hyperspectral images (HSIs) acquire hundreds of contiguous spectral bands covering the ultraviolet to infrared regions, providing substantially richer spectral information. HSIs are widely used in agricultural monitoring [2], [3], urban planning [4], [5], land-cover survey [6], and target recognition [7].

Over the past few decades, HSI classification has been a fundamental task in hyperspectral data analysis. Traditional approaches have primarily relied on manually designed feature extraction methods [8], [9], [10] combined with machine learning algorithms [11], [12]. Although these traditional approaches achieved notable success, their heavy dependence on handcrafted features and expert knowledge severely limits their generalization capabilities across diverse scenarios. Deep learning (DL) methods can learn features from raw data automatically, which effectively overcomes the limitations of manual feature extraction [13], [14]. Early research focused on spectral feature extraction networks [15], [16] and spatial feature extraction networks [17], [18], extracting deep-level spatial and spectral representations through convolutional neural networks (CNNs). However, this separate extraction method may overlook the complex relationships between spectral and spatial information. Subsequently, researchers have considered extracting joint features from both spectral and spatial dimensions through 3-D CNN [19], [20]. Nevertheless, due to the local nature of CNN receptive fields, these methods may encounter limitations in effectively modeling global contextual dependencies. With the rise of the Transformer architecture, methods based on the Transformer architecture [21], [22], [23] globally model the spatial and spectral features of HSI through self-attention. However, its quadratic time complexity will generate a relatively high computational burden when dealing with high-dimensional and hyperspectral data. Recently, researchers have also used the Mamba architecture [24], [25], [26] with linear complexity for efficient HSI global classification and feature extraction.

Those DL methods have demonstrated promising potential in extracting features for HSI classification. However, as illustrated in Fig. 1(a), most existing approaches remain

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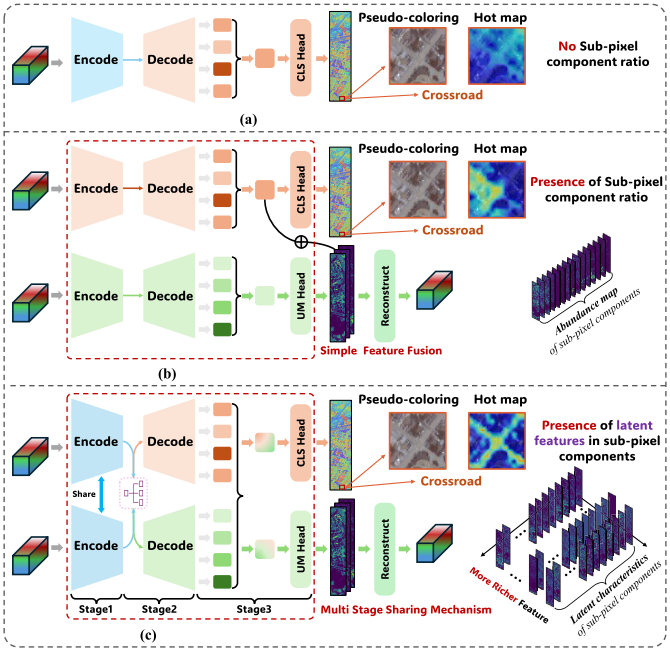


Fig. 1. Comparison of HSI classification frameworks. (a) Single-task HSI classification without subpixel component ratio estimation. (b) Multitask HSI classification with simple feature fusion, which can estimate subpixel component ratios. (c) Multitask HSI classification with multistage feature sharing, which can further extract potential subpixel component features.

constrained to a single-task learning paradigm. This isolated strategy fails to exploit the potential synergies among different tasks [27], [28] in HSI analysis. In fact, HSIs contain rich semantic information that must be interpreted from multiple task perspectives. For instance, HSI super-resolution focuses on mining subpixel spectral and spatial details [29], [30]. HSI sharpening emphasizes the enhancement of object boundary clarity [31]. HSI reconstruction aims to recover low noise and low interference spectral and spatial representations [32], and HSI unmixing is dedicated to estimating the constituent material abundances at the subpixel level [33], [34]. These tasks can provide complementary information to support classification from various dimensions. Nevertheless, efficiently sharing knowledge and features across these tasks remains a significant challenge.

Multitask learning (MTL) is a learning paradigm in which multiple related tasks [35] are trained jointly. By leveraging shared representations and cross-task knowledge transfer, MTL can effectively capture and exploit the latent relationships among tasks [36], [37]. In recent years, several studies have introduced MTL into the HSI domain and proposed various types of parameter sharing strategies. Some studies adopt a hard parameter sharing mechanism [38], [39] by constructing a backbone network that handles HSI classification and other HSI tasks simultaneously. Those methods have a simple structure and can automatically learn representations common to multiple tasks. However, since all tasks share the same parameters, it lacks the ability to model the differences between tasks, which is prone to causing interference between tasks. Other studies adopt a soft parameter sharing mechanism [40], [41] by designing a distinct subnetwork for each task and facilitating information sharing through an attention mechanism. For instance, Han et al. [42] proposed a two-branch network for unmixing and classification that delivers abundance information to the classification task via attention, thereby strengthening the cross-task semantic connection. While soft parameter sharing captures task-specific features well, its separate subnetworks limit efficient information sharing. To combine the advantages of both, some scholars have adopted a hybrid sharing mechanism. For instance, HyMuT [43] has achieved excellent performance on multiple datasets by constructing a shared backbone network and a hierarchical information transfer mechanism. However, this structure lacks the dynamic perception of task information, and it is difficult to adaptively meet task-specific requirements. In general, existing hyperspectral MTL methods can leverage auxiliary tasks to bring performance gains to classification. However, their sharing mechanisms are generally basic and lack task-specific design, failing to effectively exploit the potential complementary information and deeper correlations among different tasks.

To address these limitations, we propose a multistage information sharing multitask hyperspectral classification network with unmixing assistance (MSUANet). The proposed network employs a multistage sharing mechanism to deeply extract both shared and task-specific features from the classification and unmixing branches. Furthermore, it performs targeted information filtering tailored to the characteristics of the classification task, thereby enhancing the performance of HSI classification. MSUANet comprises three stages of information sharing. First, a shared backbone network is constructed to extract common features across tasks. Based on this, to mitigate mutual interference among tasks, we introduce an information diverter with hybrid attention (IDHA) module in the decoding stage to generate task-specific feature representations. Finally, we propose an adaptive latent abundance knowledge transfer (ALAKT) module, which dynamically perceives the importance of information generated by both classification and unmixing decoders. This module adaptively filters effective features for the classification task. As shown in Fig. 1, MSUANet leverages a multistage information sharing mechanism to make the feature activation regions of the classification task more focused on the true target areas, while more effectively exploiting the potential benefits provided by the auxiliary task. In brief, our contributions are as follows.

1) We propose an MSUANet. The network integrates shared representation learning, task-specific modeling, and adaptive information selection to fully exploit the potential of auxiliary tasks.
2) We propose an IDHA module, which explicitly models the differences between the classification and unmixing branches, enabling effective extraction of task-specific knowledge.
3) We propose an ALAKT module, which adaptively selects and transfers latent informative features from the unmixing branch to the classification task, thereby enhancing classification performance.

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The remainder of this article is organized as follows. In Section II, we introduce the relevant background knowledge of MTL and HSI unmixing tasks and the motivation of MSUANet. In Section III, we elaborate on the proposed MSUANet architecture in detail. In Section IV, we validate the proposed MSUANet through experiments conducted on four HSI datasets. The evaluation includes ablation studies, comparisons with state-of-the-art methods. Finally, Section V concludes this article and outlines potential directions for future research.

II. Background and Motivation

A. Multitask Learning

MTL jointly learns multiple related tasks instead of training individual models separately. It captures shared representations among tasks while preserving task-specific characteristics, thereby improving generalization performance across tasks. The objective of MTL is to minimize the total loss across all tasks, combining both task-specific losses and a regularization term on the model parameters. For T tasks, let θ_s denote the shared parameters and θ_j denote the task-specific parameters for task j. The optimization problem is formulated as

$$\min_{\theta_s,\theta_j} \sum_{j=1}^{T} \lambda_j \cdot \mathcal{L}_j\left(\theta_s,\theta_j\right) + \gamma \cdot \mathcal{R}\left(\theta_s,\theta_j\right) \tag{1}$$

where $\mathcal{L}_j(\theta_s,\theta_j)$ is the loss function for task j, λ_j is a weight controlling the contribution of task j, $\mathcal{R}(\cdot)$ is a regularization term applied to the parameters, and γ balances the overall regularization strength.

Recently, many researchers have introduced MTL into HSI classification by leveraging auxiliary tasks such as superpixel modeling [44], [45], spectral reconstruction [46], and domain adaptation [47], among others. These auxiliary tasks provide useful prior information, enabling the model to improve its representation learning and thus better support the primary classification task. For instance, Li et al. [48] introduced a graph-based sample relationship learning task, optimizing sample distances in the feature space via a Siamese network, effectively mitigating the overfitting problem in few-shot HSI classification. Liu et al. [49] proposed a spectral reconstruction auxiliary task, which uses reconstruction errors to detect and identify unknown classes, significantly enhancing the generalization capability of the HSI classifier in open-world scenarios. Chen et al. [50] developed a cross-domain contrastive learning task, aligning cross-domain samples in the feature space to facilitate the transfer of knowledge from the source domain to the target domain, thereby significantly improving the accuracy of HSI classification across diverse scenes.

Most existing methods enhance feature representation by directly fusing information from auxiliary tasks. However, this naive aggregation strategy often overlooks the correlations between tasks and lacks an effective mechanism for selecting useful information. Consequently, it may introduce redundant or even conflicting information [51], diminishing the benefits of auxiliary tasks. We propose a multistage sharing strategy that combines low-level feature sharing, mining of task-specific representations, and adaptive selection of task-relevant information, fully leveraging the potential of auxiliary tasks to achieve more effective MTL.

B. HSI Unmixing

HSI provides the detailed spectral information but suffers from mixed pixels due to limited spatial resolution. HSI unmixing decomposes these pixels into endmembers and abundances, revealing material composition and distribution. HSI unmixing [52] is generally categorized into linear unmixing model (LMM) and nonlinear unmixing model (NLMM) based on the mixing mechanisms of endmembers within a pixel. Under the LMM, the spectral signature of each component combines in a linearly proportional manner. In contrast, NLMM accounts for complex interactions between endmembers, resulting in mixed spectra that cannot be accurately represented through linear combinations. Here, we focus on the LMM. Let $\mathbf{Y} \in \mathbb{R}^{B\times H\times W}$ denote an HSI composed of C endmembers, with B spectral bands and spatial dimensions of $H \times W$. The LMM can be formulated as

$$\begin{aligned} &\mathbf{Y} = \mathbf{M}\mathbf{A} + \mathbf{N} \\ &\text{s.t. } \mathbf{1}_C^T\mathbf{A} = \mathbf{1}_{H\times W}^T, \quad \mathbf{A} \geq 0,\ \mathbf{M} \geq 0 \end{aligned} \tag{2}$$

where $\mathbf{M}$ is the endmember matrix, $\mathbf{A}$ is the abundance matrix, and $\mathbf{N}$ denotes the noise associated with the HSI. To ensure physical validity, the abundances are constrained to be nonnegative and sum-to-one, while the endmembers are also required to be nonnegative.

DL-based unmixing methods [53] are capable of learning more expressive representations of the spectral space. Typically, such approaches can be divided into two main steps: abundance estimation and hyperspectral reconstruction. In the abundance estimation stage, the HSI $\mathbf{Y}$ is fed into an encoder network to generate the estimated abundance $\hat{\mathbf{A}}$, which is formulated in (3). Once the abundance estimates are obtained, the HSI is reconstructed by performing a 1×1 convolution to combine the endmembers accordingly, as expressed in (4)

$$\hat{\mathbf{A}} = f_\theta\left(\mathbf{Y}\right) \tag{3}$$

$$\hat{\mathbf{Y}} = f_{\text{conv}}\left(\hat{\mathbf{A}}\right) = \mathbf{M}\hat{\mathbf{A}} \tag{4}$$

where $f_\theta(\cdot)$ denotes the abundance estimation network parameterized by weights θ and $\hat{\mathbf{A}}$ denotes the estimated abundance. The operator $f_{\text{conv}}(\cdot)$ corresponds to a 1×1 convolution whose kernel weights $\mathbf{M}$ represent the spectral signatures of the C endmembers. Consequently, $\hat{\mathbf{Y}}$ denotes the reconstructed HSI.

In recent years, HSI unmixing has been increasingly employed as an auxiliary task to enhance the performance of various HSI tasks. For instance, Zou and Huang [54] and Qu et al. [55] incorporated unmixing into HSI super-resolution, where the extracted endmembers and abundances provide spectral and spatial priors, thereby improving reconstruction quality while preserving spectral consistency. Younis et al. [56] utilized unmixing to assist anomaly detection, treating the derived endmembers and abundances as physically meaningful features that enhance the interpretability of the detection results. Seydi et al. [57] refined change regions through unmixing, leading to improved accuracy in change detection. Overall,

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unmixing not only offers physically interpretable spectral representations but also provides spatial priors that can yield additional information gain for other tasks. Motivated by this benefit, we adopt unmixing as an auxiliary task and employ a multistage information sharing strategy to fully exploit its potential in enhancing HSI classification performance.

C. Task-Specific Attention for HSI

In recent years, attention mechanisms have been widely applied in the field of HSI analysis. The core idea is to adaptively assign feature weights to emphasize discriminative information while suppressing redundant features. Early studies mainly embedded spatial and spectral attention into CNNs [58], [59] to enhance discriminative feature responses. Wu et al. [60] employed a dual-branch 3-D CNN to separately capture spatial and spectral attention features of HSIs, thereby achieving collaborative enhancement at the feature level. Li et al. [61] integrated the squeeze and excitation attention [62] with multiscale convolutions to realize adaptive recalibration of spatial and spectral hierarchical features. Chen et al. [63] incorporated the convolutional block attention [64] into CNNs, jointly modeling channel and spatial attention to further improve classification performance. With the rise of self-attention mechanisms [65], [66], Hong et al. [67] proposed SpectralFormer, which revisits HSI classification through spectral sequence modeling. By learning local dependencies among adjacent spectral bands and employing cross-layer skip connections, SpectralFormer enables adaptive fusion of multilevel features. Sun et al. [68] introduced SSFTT, which tokenizes spectral and spatial features separately before feeding them into a Transformer encoder, capturing high-order semantic relationships and significantly improving classification accuracy. Subsequently, HyperTransformer [69] fuses spectral and texture information across multiple spatial scales and models cross-modal feature dependencies to enhance representation capability.

Overall, these attention mechanisms have made significant progress in spatial spectral feature modeling. However, in hyperspectral MTL, there is a lack of exclusive feature enhancement for each task, making it difficult to screen relevant and irrelevant features for each task, which leads to the introduction of redundant and confusing information. Inspired by the self-attention mechanism, we propose the IDHA, which performs task-specific feature modeling to capture the task of exclusive information, reduce feature interference, and improve the overall expressiveness of the learned representations.

III. Methodology

MSUANet is designed to fully exploit the potential benefits of auxiliary tasks for the main classification task through a multistage information sharing strategy. As illustrated in Fig. 2, the framework consists of three information sharing stages and one task head stage. In the information sharing phase, the model sequentially performs shared feature extraction, task-specific information enhancement, and adaptive information selection, thereby progressively extracting and refining useful representations for each task. Subsequently, the processed information is delivered to their respective task processing modules. The following will provide a detailed description of the multistage sharing mechanism in MSUANet and the corresponding improvements applied to the stages.

A. Multistage Sharing Mechanism

The multistage sharing mechanism of MSUANet is shown in Fig. 2. This mechanism mainly consists of three stages: shared feature extraction, task-specific information enhancement, and adaptive information selection.

1) Shared Feature Extraction Stage: In the first stage, a unified feature encoder extracts deep representations from HSI, enabling both unmixing and classification tasks to benefit from a rich and consistent feature space. We employ Swin Transformer [70] as the backbone network to leverage its strong multiscale modeling capability and global context awareness, thus capturing rich spatial and spectral features that provide a solid foundation for subsequent task learning.

2) Task-Specific Information Enhancement Stage: The second stage focuses on task-specific feature refinement for different decoding branches. Since classification primarily emphasizes spatial semantic information, whereas unmixing focuses on spectral variation, we design the IDHA module to selectively enhance task-specific key features and boost discriminative representations for each task branch.

3) Adaptive Latent Information Selection Stage: In the third stage, we perform adaptive feature selection after task decoding. To improve feature discriminability while suppressing redundancy, we introduce the ALAKT module, which assigns importance scores to features at different levels through an adaptive weighting mechanism, selectively enhancing critical information while suppressing irrelevant or noisy features.

Through this multistage sharing strategy, MSUANet effectively leverages the information gain from the unmixing task to provide more fine-grained spectral features for classification, thereby improving classification performance. Experimental results demonstrate that this structure outperforms existing approaches by more effectively exploiting the complementary benefits of auxiliary tasks (Section IV-E).

B. Information Diverter With Hybrid Attention

In MTL frameworks, using identical decoders for different tasks is common. While this approach is convenient, it often results in significant feature overlap [51] across tasks. The shared encoder fails to explicitly model task-specific features, and homogeneous decoders lack sufficient task-specific guidance, causing intermediate representations to converge. This causes semantic ambiguity and gradient conflicts, reducing the auxiliary tasks' benefit to the primary task. However, when MTL is applied to an HSI task, these issues become even more pronounced due to the high dimensionality and complex spectral correlations inherent in HSI data. For classification tasks, discriminative spatial and spectral representations are essential for distinguishing different ground materials, whereas unmixing tasks demand the capture of fine-grained spectral variations and relationships among spectral components. Many

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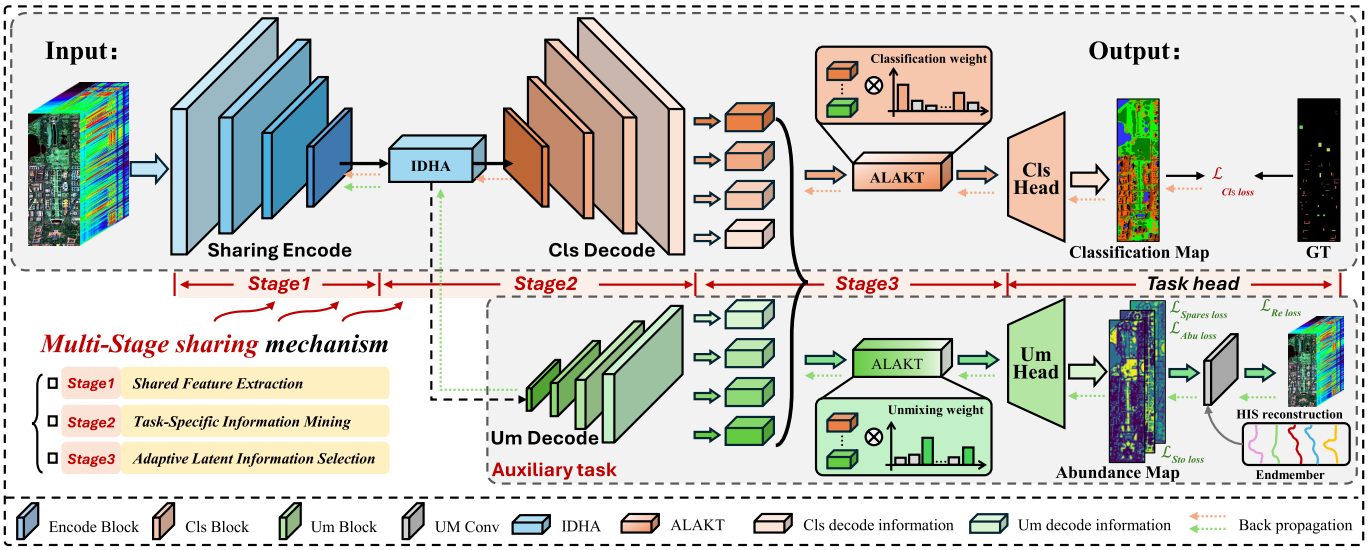


Fig. 2. Overall architecture of the proposed MSUANet, which consists of three information sharing stages followed by task-specific head stages. "Cls Decode" and "UM Decode" are the decoding modules for the classification and unmixing branches, respectively. "Cls Head" and "Um Head" serve as the prediction output layers for the classification and unmixing tasks, respectively. "IDHA" refers to the information diverter with hybrid attention module, and "ALAKT" refers to the adaptive latent abundance knowledge transfer module.

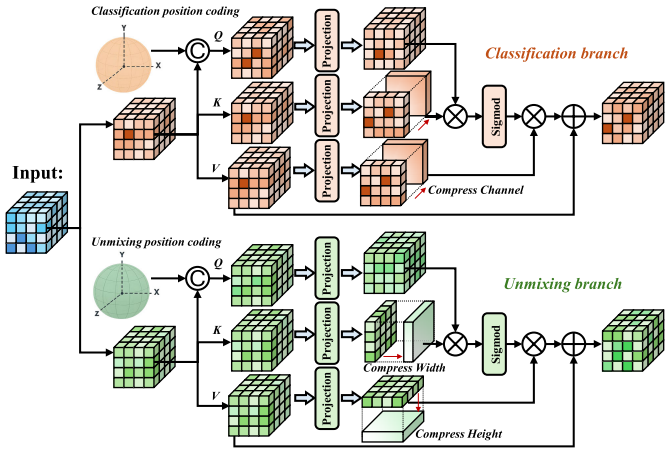


Fig. 3. Structure of the IDHA module, in which the symbol © denotes the feature concatenation, ⊗ indicates the Hadamard product, and ⊕ represents the element-wise addition.

existing methods enhance feature representations through spatial and spectral attention [60], [61], which strengthen the discriminative capacity of features and globally model spectral dependencies via self-attention [67], [68]. These approaches have achieved outstanding performance in HSI classification tasks. However, they primarily focus on improving class separability and often neglect the distinct feature modeling requirements of other tasks.

To this end, we propose an IDHA module as shown in Fig. 3. The module employs task-guided attention to differentiate features, generating task-specific information that alleviates feature overlap and improves the expressive power of feature representations. Specifically, the IDHA module consists of two stages: task-specific information encoding and task-specific information enhancement.

1) Task-Specific Information Encoding: In this stage, we perform task-specific information encoding on the input features to jointly model spatial and spectral dependencies for both classification and unmixing tasks. In HSI, each pixel corresponds to a continuous spectral signature that reflects the material composition of the scene. However, the positional encoding used in conventional Transformers is primarily designed for 2-D spatial modeling, which can only represent relative positions between pixels and fails to fully capture the continuity and correlation along the spectral dimension. To address this limitation, we introduce a 3-D positional encoding scheme that enables the model to simultaneously learn spatial and spectral relationships.

For the classification task, in order to enhance the model's ability to discriminate spatial structures and spectral distributions in HSI, we incorporate a spatially guided mapping into the encoding process. This mapping characterizes continuous spatial variations and boundary structures within the scene. Given an input feature $\mathcal{F}_{\text{in}} \in \mathbb{R}^{C\times H\times W}$, we first generate spatial coordinates along the horizontal and vertical axes. These coordinates are then nonlinearly mapped via sine and cosine functions to capture continuous and periodic spatial contextual information, resulting in the 3-D positional encoding Pos_{cls}

$$\text{Pos}_{\text{cls}} = \left[\sin\left(2\pi x_{i,j}\right), \cos\left(2\pi y_{i,j}\right), \frac{x_{i,j}+y_{i,j}}{2}\right]. \tag{5}$$

For the unmixing task, in order to more precisely represent spectral characteristics and interspectral relationships, we introduce a spectrally guided mapping into the encoding process. This design allows the model to capture spectral continuity as well as spatial and spectral coupling relationships. For the input feature $\mathcal{F}_{\text{in}} \in \mathbb{R}^{C\times H\times W}$, the positional encoding Pos_{um} is defined along the spectral dimension as follows:

$$\text{Pos}_{\text{um}} = \left[x_{i,j}, y_{i,j}, \frac{\sin\left(2\pi x_{i,j}\right)+\cos\left(2\pi y_{i,j}\right)}{2}\right] \tag{6}$$

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where $x_{i,j} = i/H-1$ and $y_{i,j} = j/W-1$ represent the normalized spatial coordinates along the horizontal and vertical axes at location $i \in \{0, \ldots, H-1\}$ and $j \in \{0, \ldots, W-1\}$.

2) Task-Specific Information Enhancement: To further refine the task-specific representation learned in the previous stage, a task-specific information enhancement module is introduced. By leveraging a hybrid attention mechanism that jointly models spatial and spectral cues, it adaptively reinforces task-dependent features and improves the distinctiveness of representations across tasks.

In classification information enhancement, the input feature $\mathcal{F}_{\text{in}}$ is first fused with the 3-D positional encoding Pos_{cls} to construct the classification query feature $\mathcal{Q}_{\text{cls}}$. Subsequently, to suppress redundant spectral components and noise, the input feature $\mathcal{F}_{\text{in}}$ is compressed along the channel dimension to generate the key feature $\mathcal{K}_{\text{cls}}$ and value feature $\mathcal{V}_{\text{cls}}$, which are employed to extract discriminative spatial responses and refine spatial information

$$\mathcal{Q}_{\text{cls}} = \text{Proj}_{C' \to C}\left(\mathcal{F}_{\text{in}} \oplus \text{Pos}_{\text{cls}}\right) \tag{7}$$

$$\mathcal{K}_{\text{cls}} = \text{Proj}_{C \to 1}\left(\mathcal{F}_{\text{in}}\right) \tag{8}$$

$$\mathcal{V}_{\text{cls}} = \text{Proj}_{C \to 1}\left(\mathcal{F}_{\text{in}}\right). \tag{9}$$

Finally, by performing element-wise weighting between the query $\mathcal{Q}_{\text{cls}}$ and the key $\mathcal{K}_{\text{cls}}$, a spatial saliency mask is generated to highlight the discriminative regions in the HSI. The value feature $\mathcal{V}_{\text{cls}}$ further refines the selected regions to enhance boundary sharpness and maintain intraregion consistency, resulting in the enhanced classification features $\mathcal{F}_{\text{clsout}}$

$$\mathcal{F}_{\text{clsout}} = \tanh\left(\mathcal{Q}_{\text{cls}} \odot \mathcal{K}_{\text{cls}}\right) \odot \mathcal{V}_{\text{cls}} + \mathcal{F}_{\text{in}} \tag{10}$$

where $\text{Proj}_{C' \to C}$ and $\text{Proj}_{C \to 1}$ denote channel-wise projections, reducing the fused feature dimension from C' to C, and compressing from C to 1, respectively. The operator $\oplus$ represents feature concatenation, $\odot$ represents the Hadamard product, and tanh denotes the hyperbolic tangent activation function. $\mathcal{K}_{\text{cls}}$ produces spatial modulation weights, whereas $\mathcal{V}_{\text{cls}}$ carries the feature content to be selectively emphasized.

In the enhancement of unmixing information, the input feature $\mathcal{F}_{\text{in}}$ is first fused with the spectral positional encoding Pos_{um} to construct the unmixing query feature $\mathcal{Q}_{\text{um}}$. Subsequently, to aggregate spectral features from different directions, the input feature is compressed along the vertical and horizontal dimensions to generate the key feature $\mathcal{K}_{\text{um}}$ and the value feature $\mathcal{V}_{\text{um}}$. The spectral features were aggregated by double view to improve the ability of the model to use local spectra

$$\mathcal{Q}_{\text{um}} = \text{Proj}_{C' \to C}\left(\mathcal{F}_{\text{in}} \oplus \text{Pos}_{\text{um}}\right) \tag{11}$$

$$\mathcal{K}_{\text{um}} = \text{Proj}_{H \to 1}\left(\mathcal{F}_{\text{in}}\right) \tag{12}$$

$$\mathcal{V}_{\text{um}} = \text{Proj}_{W \to 1}\left(\mathcal{F}_{\text{in}}\right). \tag{13}$$

Finally, the vertical spectral information $\mathcal{K}_{\text{um}}$ is employed to excite the query feature, highlighting band responses that are critical for endmember decomposition. Meanwhile, the horizontal spectral information $\mathcal{V}_{\text{um}}$ further refines and compensates these regions, yielding the enhanced unmixing feature information $\mathcal{F}_{\text{umout}}$

$$\mathcal{F}_{\text{umout}} = \tanh\left(\mathcal{Q}_{\text{um}} \odot \mathcal{K}_{\text{um}}\right) \odot \mathcal{V}_{\text{um}} + \mathcal{F}_{\text{in}} \tag{14}$$

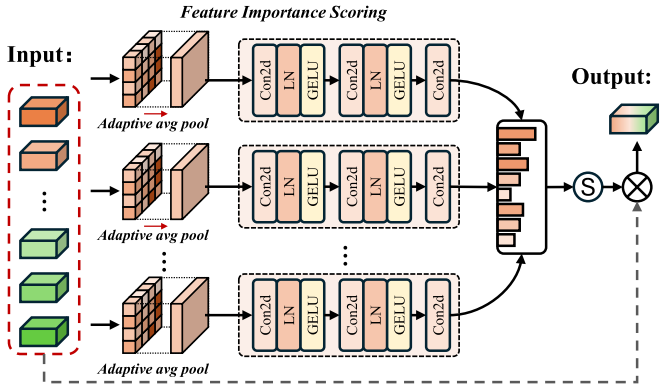


Fig. 4. Structure of the ALAKT module, in which Ⓢ denotes the Softmax operation and ⊗ indicates Hadamard product.

where $\text{Proj}_{H \to 1}$ and $\text{Proj}_{W \to 1}$ represent the compression of spatial features from H and W to 1, respectively.

C. Adaptive Latent Abundance Knowledge Transfer

In the HSI unmixing-assisted classification framework, high-level semantic features are progressively decoded via multistage upsampling to generate task-specific refined representations. For classification, these features are decoded into spatially discriminative maps highlighting land-cover differences, while for unmixing, they are transformed into latent abundance features with potential physical meaning, revealing material composition.

However, effectively leveraging abundance information from unmixing to enhance classification while reducing redundancy and interference remains challenging. To address this, we propose an ALAKT module as shown in Fig. 4, which dynamically evaluates the importance of abundance features, selectively emphasizing key information and suppressing irrelevant details to improve classification.

First, we define obtaining the high-level semantic information from the classification decode as $X_{\text{cls}}^{(l)}$ and the potential abundance information obtained from the unmixing decode as $X_{\text{um}}^{(l)}$

$$X_{\text{cls}}^{(l+1)} = \mathcal{U}\left(\mathcal{T}_{\text{cls}}^{(l)}\left(X_{\text{cls}}^{(l)}; \theta_{\text{cls}}^{(l)}\right)\right), \quad \text{for } l = 1, 2, 3 \tag{15}$$

$$X_{\text{um}}^{(l+1)} = \mathcal{U}\left(\mathcal{T}_{\text{um}}^{(l)}\left(X_{\text{um}}^{(l)}; \theta_{\text{um}}^{(l)}\right)\right), \quad \text{for } l = 1, 2, 3 \tag{16}$$

where $l = 1$, $X_{\text{cls}}^{(1)}$ and $X_{\text{um}}^{(1)}$ denote the deepest high-level semantic feature inputs for the classification and unmixing decoders, respectively. When $l = 2, 3$, intermediate features are progressively obtained through decoding and upsampling stages in each decoder. Finally, when $l = 4$, the most refined high-level feature representations are produced as the outputs from both decoders. Additionally, $\mathcal{T}_{\text{cls}}^{(l)}$ and $\mathcal{T}_{\text{um}}^{(l)}$ denote the feature decoding operation at the lth decoding layer of classification and unmixing decoders, respectively; $\theta_{\text{cls}}^{(l)}$ and $\theta_{\text{um}}^{(l)}$ are their corresponding sets of learnable parameters; and $\mathcal{U}$ indicates the upsampling operation.

Subsequently, we assign importance scores to each high-level semantic feature $X_{\text{cls}}^{(l)}$ and $X_{\text{um}}^{(l)}$ to adaptively assess their contributions to the task. Specifically, we first compute the

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channel mean response of each feature, capturing the global contextual information of the feature channels. These mean responses are then passed through a feature scoring module $\mathcal{D}_{\text{score}}$ composed of a 1×1 convolution, dynamically assigning fusion weights to each feature according to its semantic significance. The overall scoring process is expressed as

$$\mathcal{S} = \mathcal{D}_{\text{score}}\left(\bigoplus_{l=1}^{4}\left[\text{GAP}\left(X_{\text{cls}}^{(l)}\right), \text{GAP}\left(X_{\text{um}}^{(l)}\right)\right]\right) \tag{17}$$

where $\mathcal{S} \in \mathbb{R}^{1\times 8}$ is the resulting importance score vector, $\bigoplus$ denotes the multilevel feature concatenation operation, GAP$(\cdot)$ represents global adaptive pooling, and $\mathcal{D}_{\text{score}}(\cdot)$ is the feature scoring function.

Finally, we calculate the feature importance score using formula (17) and normalize the score through the Softmax function to obtain the final fusion information $\mathcal{F}_{\text{fuse}}$

$$\mathcal{F}_{\text{fuse}} = \text{Softmax}(S) \cdot \left(\sum_{l=1}^{4} X_{\text{cls}}^{(l)}, \sum_{l=1}^{4} X_{\text{um}}^{(l)}\right). \tag{18}$$

Overall, through the proposed ALAKT module, important features required by the specific task can be adaptively highlighted, while redundant and interfering features are effectively suppressed. The validity of the obtained importance weights for both unmixing and classification tasks will be further analyzed and verified in the experiments presented in Section IV-F.

D. Loss Function

The optimization objective of MSUANet comprises two tasks: classification and unmixing. These two tasks are optimized using a joint loss, where the unmixing information complements the classification task. Specifically, the classification task employs a cross-entropy loss to minimize the discrepancy between predicted feature categories and ground-truth labels. The $\mathcal{L}_{\text{cls}}$ loss can be formulated as follows:

$$\mathcal{L}_{\text{cls}} = -\frac{1}{N_t}\sum_{i=1}^{N_t} p_i \log(\hat{p}_i) \tag{19}$$

where N_t is the number of training samples, p_i is the one-hot encoding of the label, and $\hat{p}_i$ is the prediction probability of MSUANet.

In the unmixing task, the mean squared error (mse) is employed to optimize the abundance reconstruction loss $\mathcal{L}_{\text{abu}}$ and HSI reconstruction loss $\mathcal{L}_{\text{rec}}$. To ensure that the abundance maps are physically meaningful, abundance sparsity $\mathcal{L}_{\text{spa}}$ and sum-to-one constraints $\mathcal{L}_{\text{sum}}$ are also incorporated. The formulas are as follows:

$$\mathcal{L}_{\text{abu}} = \frac{1}{N_a}\sum_{i=1}^{N_a}\sum_{k=1}^{K}\left(a_{i,k} - \hat{a}_{i,k}\right)^2 \tag{20}$$

$$\mathcal{L}_{\text{rec}} = \frac{1}{N_a}\sum_{i=1}^{N_a}\sum_{k=1}^{B}\left(x_{i,b} - \hat{x}_{i,b}\right)^2 \tag{21}$$

$$\mathcal{L}_{\text{spa}} = \frac{1}{N_a}\sum_{i=1}^{N_a}\sum_{k=1}^{K}|\hat{a}_{i,k}|_* \tag{22}$$

$$\mathcal{L}_{\text{sum}} = \frac{1}{N_a}\sum_{i=1}^{N_a}\left|1 - \sum_{k=1}^{K}\hat{a}_{i,k}\right| \tag{23}$$

where $a_{i,k}$ and $\hat{a}_{i,k}$ denote the true and reconstructed abundances, respectively, of the kth endmember in the ith pixel; and $x_{i,k}$ and $\hat{x}_{i,k}$ represent the true and reconstructed HSI values, respectively, of the bth band in the ith pixel. N_a refers to the sum of the pixel values in the input HSI. K is the number of endmembers, and B denotes the number of spectral bands in the image. $|\cdot|_*$ represents the kernel function that enforces sparsity constraints, while $|\cdot|$ denotes the $L1$ norm regularization.

To prevent training instability arising from scale differences among loss terms and varying task difficulties during joint multitask optimization, we adopt a dynamic weighting [71] method based on task uncertainty, which automatically adjusts the loss weights for each task. The final loss $\mathcal{L}_{\text{total}}$ is

$$\begin{aligned}\mathcal{L}_{\text{total}} = &\frac{1}{2\sigma_1^2}\cdot\mathcal{L}_{\text{cls}} + \frac{1}{2\sigma_2^2}\cdot\mathcal{L}_{\text{abu}} + \frac{1}{2\sigma_3^2}\cdot\mathcal{L}_{\text{rec}} \\ &+ \frac{1}{2\sigma_4^2}\cdot\mathcal{L}_{\text{spa}} + \frac{1}{2\sigma_5^2}\cdot\mathcal{L}_{\text{sum}} + \frac{1}{2}\log\left(\sigma_1^2\sigma_2^2\sigma_3^2\sigma_4^2\sigma_5^2\right)\end{aligned} \tag{24}$$

where σ_1^2, σ_2^2, σ_3^2, σ_4^2, and σ_5^2 denote the noise variances corresponding to classification, abundance reconstruction, HSI reconstruction, abundance sparsity, and abundance sum-to-one constraint, respectively, which are optimized together with the network via backpropagation. $\log(\sigma_1^2\sigma_2^2\sigma_3^2\sigma_4^2\sigma_5^2)$ is introduced as a balancing factor to regulate the MTL loss and prevent excessive task imbalance.

IV. Experiment and Result

In this section, we conduct a series of experiments to validate the effectiveness of the MSUANet. Section IV-A presents the dataset configurations and experimental settings. Section IV-B compares the performance of MSUANet with mainstream methods. Section IV-C evaluates performance under varying training ratios. Section IV-D presents the parameter sensitivity analysis of MSUANet. Section IV-E provides a visual analysis comparing MSUANet with other MTL architectures. Section IV-F explores the attention mechanisms involved in classification and unmixing tasks. Section IV-G reports on the ablation studies. Section IV-H analyzes the computational complexity of the model. Section IV-I reports the ablation study of the IDHA. Section IV-J reports the ablation study of the ALAKT. Section IV-K analyzes the reliability of the extracted endmembers. Finally, Section IV-L examines the model's limitations and robustness.

A. Dataset Description and Experimental Parameter Setting

To verify the effectiveness of MSUANet, we conducted experimental verification in Indian Pines, Houston 2013, Berlin, and Washington DC MALL. The details of the dataset are as follows.

1) *Indian Pines*: An airborne HSI was collected in the Indiana Pine Forest in Indiana, USA, using the AVIRIS

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TABLE I

TRAINING AND TESTING SAMPLES FOR EACH CLASS OF THE INDIAN PINES

Class	Name	Train	Test	All
1	Alfalfa	5	41	46
2	Corn-notill	5	1423	1428
3	Corn-mintill	5	825	830
4	Corn	5	232	237
5	Grass-pasture	5	478	483
6	Grass-trees	5	725	730
7	Grass-pasture-mowed	5	23	28
8	Hay-windrowed	5	473	478
9	Oats	5	15	20
10	Soybean-notill	5	967	972
11	Soybean-mintill	5	2450	2455
12	Soybean-clean	5	588	593
13	Wheat	5	200	205
14	Woods	5	1260	1265
15	Buildings-Grass-Trees-Drives	5	381	386
16	Stone-Steel-Towers	5	88	93
Total	-	80	10169	10249

TABLE II

TRAINING AND TESTING SAMPLES FOR EACH CLASS OF THE HOUSTON 2013

Class	Name	Train	Test	All
1	Healthy grass	20	1231	1251
2	Stressed grass	20	1234	1254
3	Synthetic grass	20	677	697
4	Trees	20	1224	1244
5	Soil	20	1222	1242
6	Water	20	305	325
7	Residential	20	1248	1268
8	Commercial	20	1224	1244
9	Road	20	1232	1252
10	Highway	20	1207	1227
11	Railway	20	1215	1235
12	Parking Lot 1	20	1213	1233
13	Parking Lot 2	20	449	469
14	Tennis Court	20	408	428
15	Running Track	20	640	660
Total	-	300	14729	15029

sensor, featuring rich forest and vegetation characteristics. Its spatial resolution is 20 m, containing 145 × 145 pixels and 220 spectra, with a total of 16 types of ground features. Table I shows the selected training and validation samples from the Indian Pines.

2) *Houston 2013*: An airborne HSI was collected near the University of Houston using the ITRES CASI-1500 sensor, featuring rich urban characteristics. Its spatial resolution is 2.5 m, containing 305 × 1905 pixels and 155 spectra, with a total of 15 types of ground features. Table II shows the selected training and validation samples from the Houston 2013.

3) *Berlin*: A spaceborne EnMAP simulation image, with data collection scenes in Berlin, Germany, and its surrounding areas, featuring rich urban and rural characteristics. Its spatial resolution is 30 m, containing 1723 × 476 pixels and 244 spectra, with a total of 8 types of ground features. Table III shows the selected training and validation samples from the Berlin.

TABLE III

TRAINING AND TESTING SAMPLES FOR EACH CLASS OF BERLIN

Class	Name	Train	Test	All
1	Forest	50	54904	54954
2	Residential Area	50	268592	268642
3	Industrial Area	50	19516	19566
4	Low Plants	50	59232	59282
5	Soil	50	17376	17426
6	Allotment	50	13255	13305
7	Commercial Area	50	24774	24824
8	Water	50	6622	6672
Total	-	400	464271	464671

TABLE IV

TRAINING AND TESTING SAMPLES FOR EACH CLASS OF THE WASHINGTON DC MALL

Class	Name	Train	Test	All
1	Roof	5	3829	3834
2	Road	5	411	416
3	Meadow	5	170	175
4	Tree	5	1923	1928
5	Path	5	400	405
6	Water	5	1219	1224
7	Shadow	5	92	97
Total	-	35	8074	8079

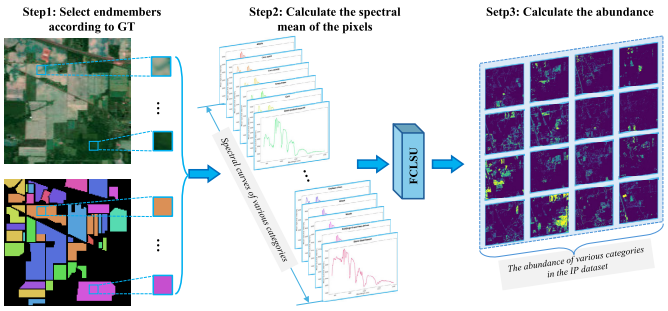


Fig. 5. Abundance and endmember label generation process for HSI unmixing.

4) *Washington DC MALL*: An airborne HSI collected over a shopping center in Washington, DC, using the HYDICE sensor, featuring a distinct urban–rural distinction. Its spatial resolution is 2.8 m, containing 1280 × 307 pixels and 191 spectra, with a total of 7 types of ground features. Table IV shows the selected training and validation samples from the Washington DC MALL.

Since HSI classification datasets lack endmember and abundance labels, we referred to CycuNet [53] to create coarse-grained unmixing labels to assist classification, and the specific process is shown in Fig. 5. First, we select a region for each category based on the ground truth. Then, we take the average spectral value as the endpoint label of this category. Finally, in order to obtain the abundance information, we use the fully constrained least-squares unmixing (FCLSU) algorithm to extract abundance based on the selected endmembers.

To quantitatively analyze the superiority of the model's performance, we used three indicators, namely, overall accuracy (OA), average accuracy (AA), and kappa coefficient (Kappa), for measurement. The experimental hardware setup

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consists of an RTX 4090 D GPU with 24-GB memory, CUDA version 11.3, and PyTorch 1.11.0. To ensure the fairness of the comparative experiments, we adopt the same experimental configuration for all patch-free methods, while for patch-based methods, we use the optimal configuration provided by the official code. Additionally, to mitigate randomness in the experiments, we conduct ten independent trials for each method and report their mean values along with standard deviations.

B. Comparison With State-of-the-Art Methods

To evaluate the effectiveness of our proposed MSUANet method, we compare it with several state-of-the-art approaches on four HSI datasets. These include the classical segmentation networks UNet [72] and UperNet [73]; HSI specific networks such as FreeNet [74], FContNet [75], and SSFCN [76]; and recent mainstream segmentation models like Segformer [77] and TransUNet [78]. As well as HSI unmixing-assisted classification methods DSNet [42] and SV2Net [41]. The details of the comparison methods are as follows.

1) *UNet [72]:* This model is a classical image segmentation network with a symmetric encoder–decoder architecture. It utilizes skip connections to integrate shallow features with deep semantic information, enhancing segmentation accuracy. The training is conducted for 300 epochs with a batch size of 1 using the Adam optimizer. The learning rate and weight decay are set to $1e^{-3}$ and $1e^{-4}$, respectively, and the optimization strategy follows a cosine annealing schedule.
2) *UperNet [73]:* This model is a classical multitask segmentation network that incorporates the feature pyramid network and pyramid pooling module to efficiently extract multiscale semantic information from images. The training is conducted for 300 epochs with a batch size of 1 using the Adam optimizer. The learning rate and weight decay are set to $1e^{-3}$ and $1e^{-4}$, respectively, and the optimization strategy follows a cosine annealing schedule.
3) *FreeNet [74]:* This model is a patch-free HSI classification network, built upon the UNet architecture with enhancements. It employs an encoder with spectral attention mechanisms to efficiently capture global spatial information. The training is conducted for 300 epochs with a batch size of 1 using the Adam optimizer. The learning rate and weight decay are set to $1e^{-3}$ and $1e^{-4}$, respectively, and the optimization strategy follows a cosine annealing schedule.
4) *FContNet [75]:* This model is a patch-free HSI classification network, which obtains the dependencies between multiscale features through the self-attention mechanism and effectively extracts the context information in the HSI classification process. The training is conducted for 300 epochs with a batch size of 1 using the Adam optimizer. The learning rate and weight decay are set to $1e^{-3}$ and $1e^{-4}$, respectively, and the optimization strategy follows a cosine annealing schedule.
5) *SSFCN [76]:* This model is a patch-free HSI classification network, which can efficiently extract the ground object attributes of HSI through end-to-end training of spectral features and spatial features. The training is conducted for 300 epochs with a batch size of 1 using the Adam optimizer. The learning rate and weight decay are set to $1e^{-3}$ and $1e^{-4}$, respectively, and the optimization strategy follows a cosine annealing schedule.
6) *Segformer [77]:* This model is an efficient Transformer-based semantic segmentation network, which outputs multiscale feature information through a hierarchical Transformer encoder. A lightweight decoder is used to aggregate the information of different layers, which has achieved efficient segmentation. The training is conducted for 300 epochs with a batch size of 1 using the Adam optimizer. The learning rate and weight decay are set to $1e^{-3}$ and $1e^{-4}$, respectively, and the optimization strategy follows a cosine annealing schedule.
7) *TransUNet [78]:* This model is an efficient semantic segmentation network based on the combination of Transformer and CNN. It extracts CNN features efficiently through the Transformer encoder and uses the cross-attention mechanism to refine the feature information, which has achieved efficient semantic segmentation. The training is conducted for 300 epochs with a batch size of 1 using the Adam optimizer. The learning rate and weight decay are set to $1e^{-3}$ and $1e^{-4}$, respectively, and the optimization strategy follows a cosine annealing schedule.
8) *DSNet [42]:* This model is a two-branch unmixing guided classification network, which generates abundance information through an autoencoder to assist the classification branch to make efficient decisions. The training is performed for 500 epochs with a batch size of 64. The Adam optimizer is used with a learning rate of $1e^{-3}$, and the StepLR strategy is applied for learning rate decay.
9) *SV2Net [41]:* This model is a modified network of DSNet, where the variability of spectra is further considered based on DSNet to ensure the construction of accurate endmembers. The training is performed for 500 epochs with a batch size of 64. The Adam optimizer is used with a learning rate of $1e^{-3}$, and the StepLR strategy is applied for learning rate decay.

It is worth noting that MSUANet adopts a patch-free paradigm. Unlike patch-based paradigms that divide HSI into local patches for processing, it directly operates on the entire image, reducing the redundant computational part caused by dividing patches. Overall, this paradigm difference has a relatively small impact on the experimental results. Because we use the best settings from the original article for the patch-based method to ensure the best results. Tables V–VIII show the comparison results between our method and mainstream methods, and Figs. 6–9 show the classification map:

1) Indian Pines: The comparative results on the Indian Pines dataset are presented in Table V. Under limited-sample training conditions, both UNet and SSFCN exhibit suboptimal classification performance, struggling to effectively capture the fine-grained features inherent in HSI. In contrast, DSNet and SV2Net, which incorporate an unmixing task, achieve

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TABLE V
CLASSIFICATION EFFECTS OF VARIOUS METHODS ON THE INDIAN PINES DATASET

Model	UNet [72]	FreeNet [74]	FContNet [75]	SSFCN [76]	UperNet [73]	Segformer [77]	TransUNet [78]	DSNet [42]	SV2Net [41]	Ours
Class 1	65.61±23.05	88.05±20.79	85.61±10.77	0.49±0.98	99.51±0.98	56.59±10.39	71.95±12.54	44.65±9.86	60.00±11.46	**99.76±0.73**
Class 2	18.45±9.93	32.54±5.81	43.97±8.38	3.93±6.29	48.19±5.75	27.24±8.56	31.71±9.09	36.25±13.18	**56.50±6.33**	44.49±8.45
Class 3	11.20±4.82	51.37±7.92	36.85±4.95	2.61±4.63	48.56±6.48	34.79±6.55	38.21±9.62	63.83±10.51	**72.70±14.51**	55.99±11.08
Class 4	12.24±8.55	81.68±7.11	81.08±4.70	1.90±3.77	**91.72±2.72**	58.23±16.37	70.00±10.32	64.02±5.25	73.40±7.12	86.90±10.28
Class 5	29.48±14.78	70.02±7.49	68.08±6.31	9.12±18.94	78.08±5.16	62.76±8.94	56.90±8.07	71.94±6.91	**90.30±5.50**	86.40±4.19
Class 6	27.92±15.01	42.19±8.38	52.26±9.43	5.82±7.78	51.24±8.65	38.70±10.71	44.36±7.27	88.56±6.26	**96.20±3.49**	89.24±6.27
Class 7	46.96±27.14	97.83±5.24	**100.00±0.00**	3.91±7.89	**100.00±0.00**	93.91±11.20	85.22±28.92	47.47±4.65	62.40±9.43	**100.00±0.00**
Class 8	57.51±23.49	85.12±8.63	97.59±2.05	12.33±17.71	97.67±4.89	84.78±9.84	81.14±14.62	60.90±5.60	72.70±4.97	**99.96±0.08**
Class 9	89.33±29.84	98.67±2.67	**100.00±0.00**	18.67±34.62	**100.00±0.00**	96.00±5.33	**100.00±0.00**	26.62±4.36	37.90±4.56	**100.00±0.00**
Class 10	31.53±18.01	56.22±14.03	61.94±10.72	1.37±3.76	75.90±9.84	42.02±11.11	57.52±14.00	75.32±12.15	**97.40±4.30**	77.27±9.87
Class 11	27.56±26.79	56.05±11.42	53.02±6.67	85.67±11.03	63.28±8.98	51.53±8.78	53.99±15.78	89.89±5.75	**90.00±7.69**	83.15±8.12
Class 12	13.52±6.93	45.68±7.17	39.54±7.79	6.53±11.11	61.85±11.22	29.35±17.54	33.81±7.54	38.52±7.39	68.10±9.93	**69.25±6.91**
Class 13	56.65±24.73	92.50±4.69	**99.80±0.46**	12.75±24.42	89.40±5.29	82.50±14.01	95.65±5.44	92.32±6.51	93.80±4.13	96.20±2.68
Class 14	40.82±17.31	83.36±4.30	80.69±3.13	84.83±28.87	85.59±3.04	80.91±4.44	62.62±10.11	82.56±12.69	**96.00±2.49**	90.06±2.78
Class 15	17.48±9.09	88.71±6.52	87.74±5.04	9.74±16.39	93.81±4.26	68.87±5.64	70.29±8.47	89.61±12.75	**100.00±0.00**	97.06±2.88
Class 16	87.39±29.29	90.45±7.47	97.73±3.63	31.82±34.98	98.75±2.00	88.52±8.11	91.02±5.17	**100.00±0.00**	**100.00±0.00**	98.18±2.05
OA	28.33±2.64	59.31±2.95	60.22±2.59	34.81±3.34	68.19±1.11	51.30±3.06	52.87±3.60	59.28±2.96	71.80±1.81	**77.52±1.67**
AA	39.60±9.77	72.53±1.99	74.12±1.88	18.22±4.59	80.22±0.76	62.29±3.24	65.27±3.36	67.03±2.94	79.30±1.34	**85.87±1.04**
Kappa	20.71±2.65	54.36±3.10	54.92±3.00	21.64±3.91	64.21±1.06	44.86±3.42	46.88±3.44	54.02±3.32	67.80±1.99	**74.43±1.83**

TABLE VI
CLASSIFICATION EFFECTS OF VARIOUS METHODS ON THE HOUSTON 2013

Model	UNet [72]	FreeNet [74]	FContNet [75]	SSFCN [76]	UperNet [73]	Segformer [77]	TransUNet [78]	DSNet [42]	SV2Net [41]	Ours
Class 1	91.11±2.77	91.80±4.12	**98.05±1.05**	82.32±4.83	86.81±3.05	88.62±6.11	88.38±4.32	97.13±1.12	97.58±2.39	94.40±4.48
Class 2	87.26±3.35	86.50±3.84	95.88±2.46	92.51±3.56	79.86±6.22	82.01±3.83	92.79±4.68	98.14±0.51	**98.30±1.58**	93.39±2.87
Class 3	**99.97±0.09**	99.86±0.11	99.78±0.60	94.60±3.47	99.49±0.61	93.76±8.37	98.08±3.91	99.83±0.23	99.85±0.28	99.86±0.06
Class 4	96.59±1.52	92.70±2.23	94.52±1.68	81.30±8.26	86.76±6.19	64.28±12.39	94.37±3.03	96.19±1.98	97.83±1.57	**98.55±1.12**
Class 5	98.83±0.93	97.02±2.23	**99.27±0.36**	93.11±5.06	95.80±2.53	91.08±4.55	95.66±2.82	99.16±0.66	98.94±1.11	99.21±0.39
Class 6	96.69±2.82	98.19±0.86	99.25±1.08	93.09±3.03	98.12±0.00	89.31±3.43	96.28±3.37	74.73±1.29	84.82±1.45	**99.97±0.09**
Class 7	75.99±4.04	85.31±3.93	82.53±2.32	61.27±8.71	83.41±4.53	58.94±5.40	81.88±4.28	85.64±1.64	**91.31±2.76**	82.41±3.09
Class 8	64.02±10.01	76.70±5.44	81.77±5.34	54.52±8.45	63.34±6.56	45.81±8.48	78.54±4.51	64.52±6.19	69.02±4.10	**86.59±3.30**
Class 9	76.51±10.74	76.44±6.06	84.86±2.01	71.34±4.89	73.58±7.30	73.17±5.67	85.19±3.42	79.76±1.52	**85.81±2.89**	85.36 ±5.93
Class 10	69.21±7.33	85.13±8.41	85.97±2.17	61.09±9.06	90.82±2.97	68.27±14.77	89.93±10.38	90.89±5.08	93.87±3.51	**98.63±2.51**
Class 11	79.41±3.20	75.89±2.30	83.59±2.65	46.59±7.58	77.42±1.91	33.91±4.46	73.41±5.77	80.35±4.98	**88.42±4.42**	80.33±5.26
Class 12	86.77±8.81	91.05±3.26	89.80±3.96	54.75±7.28	89.67±3.77	51.28±8.38	88.66±6.82	86.87±3.19	93.79±3.14	**97.90±0.85**
Class 13	95.32±1.37	95.58±2.38	**97.50±2.29**	38.69±4.95	96.57±2.31	60.65±3.72	94.07±4.68	46.99±3.77	91.96±3.92	93.17±3.16
Class 14	99.98±0.07	99.48±0.49	**100.00±0.00**	92.74±3.41	92.70±10.64	88.94±5.64	99.57±0.70	99.82±0.38	99.90±0.31	99.65±0.24
Class 15	**99.98±0.05**	98.70±1.38	98.79±0.75	94.95±4.23	99.77±0.44	94.99±3.99	98.61±1.96	98.92±1.10	99.78±0.30	99.95±0.14
OA	85.33±1.94	88.01±1.33	91.24±0.85	72.34±2.93	85.29±1.10	69.35±2.50	88.69±1.34	77.10±12.02	92.34±0.75	**92.84±1.05**
AA	87.84±1.53	90.02±1.18	92.77±0.75	74.19±2.56	87.61±1.28	72.33±2.50	90.36±0.99	80.91±10.05	92.75±0.66	**93.96±0.83**
Kappa	84.17±2.09	87.06±1.43	90.54±0.91	70.11±3.17	84.11±1.19	66.92±2.70	87.80±1.44	86.70±0.94	91.72±0.81	**92.27±1.13**

significantly better classification accuracy, suggesting that the inclusion of unmixing enhances the model's discriminative capability. In addition, our method is significantly better than other mainstream methods in terms of OA (77.52%), AA (85.87%), Kappa (74.43%), and other indicators. Compared to the second-best method, SV2Net, our model yields improvements of +5.72%, +6.57%, and +6.63% in OA, AA, and Kappa, respectively.

As shown in Fig. 6, in the large block soybean-mintill category, most of the models have shown a lot of noise. Especially, the UNet has almost no discrimination ability, while our MSUANet only shows a small amount of noise.

2) Houston 2013: The comparative results on the Houston 2013 dataset are presented in Table VI, showing that under the 20 classification categories, most models achieved accuracies ranging from 80% to 90%. However, it is notable that Segformer, based on a Transformer architecture, exhibited the poorest performance, even significantly underperforming the traditional UNet model. This inferior performance could be attributed to Segformer's simplistic MLP structure in the decoding stage, hindering its ability to effectively recover detailed HSI information. In contrast, the proposed MSUANet demonstrated superior performance, clearly outperforming other mainstream methods in terms of OA (92.84%), AA (93.96%), and Kappa (92.27%). Furthermore, MSUANet consistently achieved classification accuracies above 82% across all categories, particularly excelling in key classes such as Classes 6, 10, and 12.

As shown in Fig. 7, Segformer exhibits noticeable noise and chaotic classification within regions, resulting in blurred boundaries between different land-cover classes. In contrast, our proposed MSUANet generates significantly fewer noisy

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TABLE VII
CLASSIFICATION EFFECTS OF VARIOUS METHODS ON BERLIN

Model	UNet [72]	FreeNet [74]	FContNet [75]	SSFCN [76]	UperNet [73]	Segformer [77]	TransUNet [78]	DSNet [42]	SV2Net [41]	Ours
Class 1	53.05±22.17	80.35±4.83	83.62±4.67	80.40±1.82	53.80±20.04	67.72±5.06	72.01±9.42	**84.80±1.96**	77.35±12.34	72.78±16.67
Class 2	76.86±13.83	72.96±5.11	72.63±3.27	69.58±9.83	85.68±8.45	73.84±5.80	76.44±8.97	72.25±5.90	78.26±4.92	**85.76±7.90**
Class 3	14.59±16.30	56.94±15.27	**69.02±4.42**	52.75±8.61	25.31±18.47	64.05±8.69	36.78±14.72	37.30±9.67	42.01±12.05	57.36±27.08
Class 4	61.31±20.52	81.70±4.71	**84.92±5.40**	75.52±4.44	48.02±25.90	81.80±6.35	60.46±16.25	75.64±8.95	73.14±8.76	78.44±11.24
Class 5	80.51±15.76	92.38±1.70	**94.02±2.58**	90.00±3.17	73.89±17.42	82.95±12.15	79.24±25.10	90.27±3.67	90.83±2.32	79.21±12.85
Class 6	27.54±20.12	50.97±14.71	52.08±9.20	45.41±18.14	18.48±16.83	31.23±19.15	49.01±18.67	51.81±18.68	**53.94±9.58**	34.96±26.85
Class 7	29.11±11.76	47.18±9.42	**56.88±5.69**	42.51±11.88	40.44±19.81	32.20±12.34	56.34±17.60	47.31±7.16	48.67±9.52	47.67±13.33
Class 8	49.96±34.45	87.53±3.42	**89.96±2.95**	80.85±7.63	62.67±26.19	68.47±11.82	71.65±15.03	82.16±2.70	82.96±5.80	75.59±10.03
OA	65.23±7.22	73.20±1.74	74.97±1.54	69.70±4.77	69.45±3.56	70.54±2.20	70.38±2.70	71.60±2.63	74.24±2.46	**78.22±2.78**
AA	49.12±7.48	71.25±4.34	**75.39±1.19**	67.13±3.36	51.04±11.45	62.78±5.67	62.74±7.49	67.69±2.41	68.40±2.26	66.47±7.84
Kappa	46.67±6.65	61.51±1.90	64.11±1.64	56.93±4.58	49.32±8.85	56.70±2.40	56.17±2.79	59.00±2.55	61.75±2.96	**65.54±3.71**

TABLE VIII
CLASSIFICATION EFFECTS OF VARIOUS METHODS ON THE WASHINGTON DC MALL DATASET

Model	UNet [72]	FreeNet [74]	FContNet [75]	SSFCN [76]	UperNet [73]	Segformer [77]	TransUNet [78]	DSNet [42]	SV2Net [41]	Ours
Class 1	87.87±10.67	97.43±1.34	97.62±1.31	97.49±1.33	82.67±9.72	98.21±1.12	93.04±5.55	72.79±11.80	57.44±9.10	**99.03±0.74**
Class 2	57.13±22.11	90.33±7.69	69.15±14.70	84.95±11.17	57.92±39.88	59.10±17.78	77.51±12.24	**98.39±0.61**	98.18±2.23	93.22±7.30
Class 3	65.04±32.35	93.84±8.89	87.20±9.47	60.24±20.47	67.68±37.19	67.36±25.10	68.72±31.47	93.06±17.28	**99.35±0.70**	87.60±11.12
Class 4	95.89±4.54	99.22±2.21	100.00±0.00	99.73±0.33	97.93±1.79	99.89±0.34	98.86±2.09	94.76±6.17	98.19±4.29	**99.92±0.24**
Class 5	47.01±35.04	92.59±15.28	99.30±1.63	95.69±4.07	42.48±39.73	93.35±5.91	91.89±16.44	97.40±0.61	**99.35±0.81**	98.23±3.01
Class 6	93.97±13.70	96.44±3.70	95.03±2.21	98.88±1.10	95.72±2.91	91.63±3.89	96.31±1.95	**99.49±0.29**	97.72±4.92	99.71±0.33
Class 7	76.38±28.57	**98.72±2.72**	98.72±3.18	79.57±12.59	72.77±32.82	83.62±13.82	84.26±28.25	87.07±20.29	93.04±5.23	71.28±19.34
OA	86.98±7.20	97.11±1.44	96.37±0.35	96.86±0.82	85.04±2.59	94.96±0.65	93.72±2.60	85.21±4.75	84.76±8.91	**98.68±0.44**
AA	74.76±10.84	**95.51±3.23**	92.43±2.06	88.08±4.03	73.88±18.27	84.74±4.01	87.23±6.58	91.85±4.88	88.28±5.10	92.71±3.34
Kappa	81.35±9.61	95.74±2.11	94.64±0.51	95.33±1.21	78.54±3.58	92.43±1.00	90.90±3.56	80.01±6.03	72.64±4.75	**98.04±0.65**

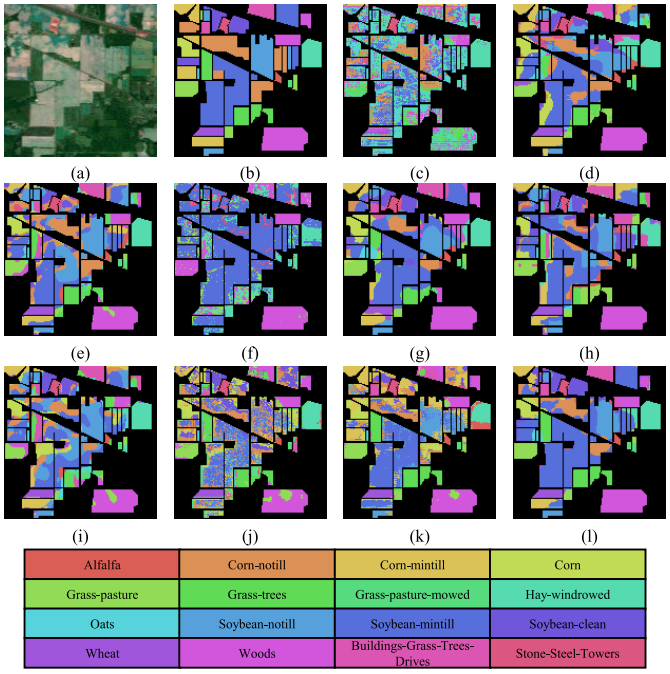


Fig. 6. Classification map on Indian Pines. (a) False-color Image. (b) GT. (c) UNet. (d) FreeNet. (e) FContNet. (f) SSFCN. (g) UperNet. (h) Segformer. (i) TransUNet. (j) DSNet. (k) SV2Net. (l) Ours.

predictions and accurately distinguishes between categories such as Parking Lot 2, Running Track, and Healthy Grass, demonstrating stronger discriminative capability and spatial consistency.

3) Berlin: The comparative results on the Berlin dataset are presented in Table VII. The Berlin dataset represents a challenging hyperspectral classification scenario, with most mainstream approaches achieving overall accuracies ranging between 60% and 75% when limited to 50 labeled samples per class. Our proposed method achieves the best results in terms of OA (78.22%) and Kappa (65.54%), surpassing the second-best method, FContNet, by margins of +3.25% and +1.43%, respectively. Although our AA is slightly lower than that of FContNet, this discrepancy is mainly due to the imbalanced distribution of class samples. In particular, Class 2, which refers to Residential Area, accounts for nearly half of the entire dataset. This causes our model to focus more on accurately recognizing dominant and large-area classes. Conversely, relatively lower accuracies are observed for minority classes, such as Class 3 Industrial Area and Class 6 Allotment, likely due in part to potential inaccuracies or spatial misalignments inherent in the original ground-truth labels directly annotated from OpenStreetMap.

As shown in Fig. 8, we observed that on Class 2 Residential Area, the most abundant category, MSUANet produces results closely aligned with the ground-truth labels, exhibiting only minor noise. In contrast, other methods introduce significant noise. For instance, the second-best performing method, FContNet, shows substantial overlapping between Class 6 Commercial Area and Class 2 Residential Area.

4) Washington DC MALL: The comparative results on the Washington DC MALL dataset are presented in Table VIII. We found that the classification networks FreeNet, FContNet, and

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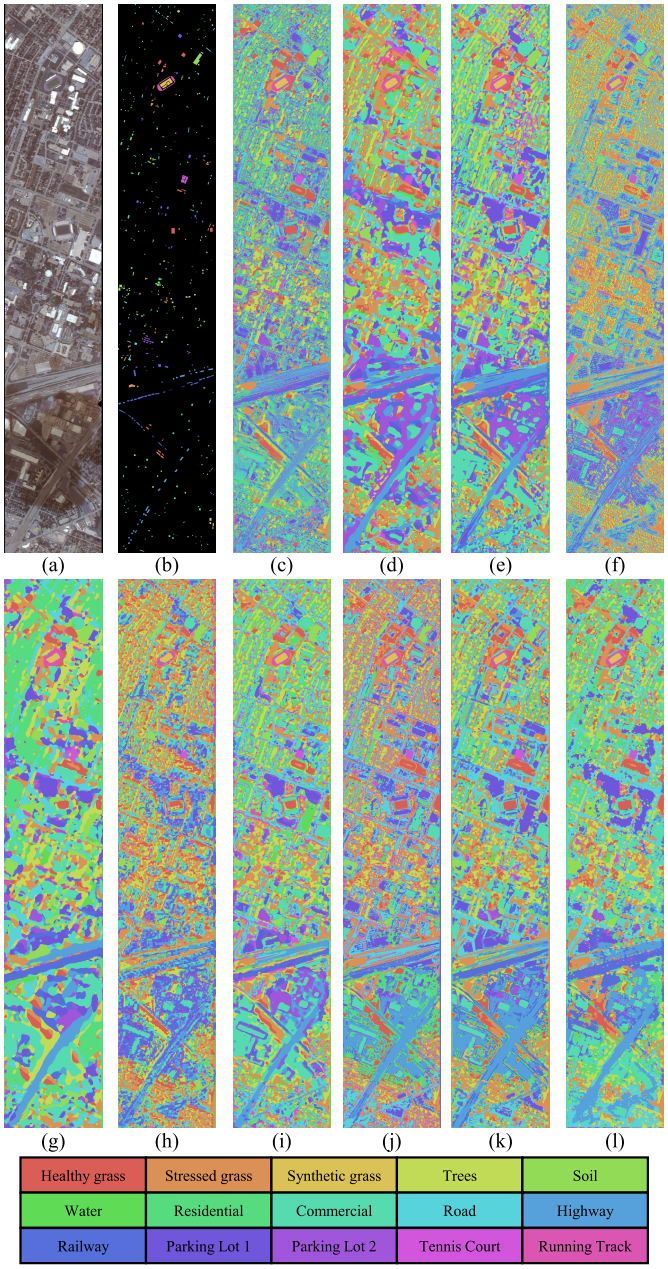


Fig. 7. Classification map on Houston 2013. (a) False-color Image. (b) GT. (c) UNet. (d) FreeNet. (e) FContNet. (f) SSFCN. (g) UperNet. (h) Segformer. (i) TransUNet. (j) DSNet. (k) SV2Net. (l) Ours.

SSFCN designed based on whole graph HSIs and Segformer and TransUNet based on Transformer architecture all exceed 93% in indicators such as OA, AA, and Kappa. However, the networks UNet and UperNet with CNN architecture and DSNet and SV2Net with the added unmixing task performed generally. This might be due to the insufficient joint extraction ability of spatial–spectral features by the CNN architecture network, and the network for the unmixing task might fail to mine the deep-level unmixing feature representations. Furthermore, compared to the suboptimal-performing FreeNet, the proposed MSUANet achieves improvements of +1.57% and +2.3% in OA and Kappa metrics, respectively, demonstrating a stronger overall classification capability. Although it slightly

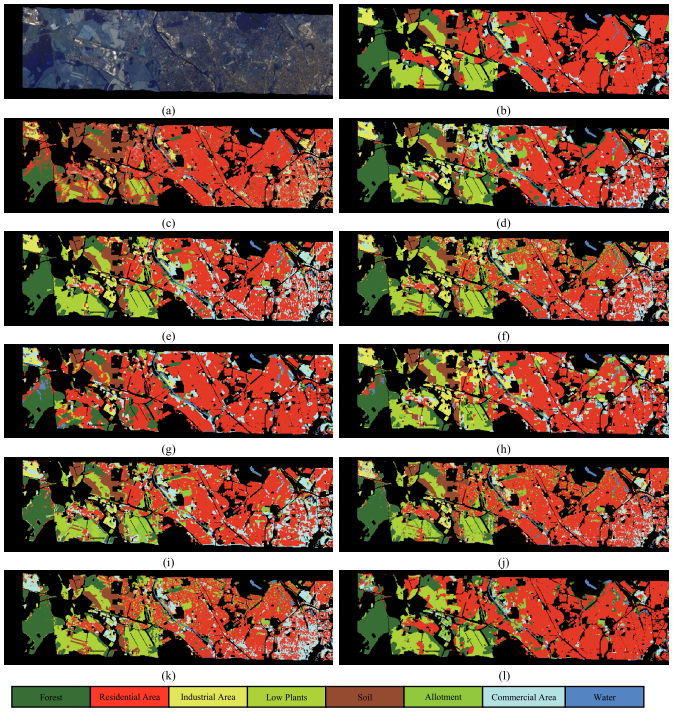


Fig. 8. Classification map on Berlin. (a) False-color Image. (b) GT. (c) UNet. (d) FreeNet. (e) FContNet. (f) SSFCN. (g) UperNet. (h) Segformer. (i) TransUNet. (j) DSNet. (k) SV2Net. (l) Ours.

underperforms FreeNet in terms of the AA metric, this may be attributed to MSUANet's tendency to capture dominant class features at the whole-image level, which can result in relatively insufficient attention to minority classes in imbalanced class distribution scenarios.

As shown in Fig. 9, we observed that the UpNet model exhibited significant misclassification in most regions, incorrectly categorizing large areas as the Water class. Although the SSFCN model demonstrated robust boundary delineation, it suffered from considerable noise artifacts. Despite the FContNet model showing visually appealing overall performance, it incorrectly classified the tree-covered regions entirely as Soil. In contrast, our proposed method successfully identified these regions as Class 4 Tree, demonstrating superior classification accuracy and reliability.

Experimental results on four representative hyperspectral datasets, including Indian Pines, Houston 2013, Berlin, and Washington DC MALL, demonstrate that the proposed MSUANet significantly outperforms existing mainstream methods in overall classification performance. This superior performance primarily stems from the proposed multistage information sharing multitask framework, which introduces spectral unmixing as an auxiliary task and leverages shared encoders along with task-specific decoders to enable deep cross-task feature interaction and collaborative representation learning.

C. Performance With Varying Numbers of Training Samples

In this section, we evaluate the generalization capabilities of the proposed MSUANet model compared to mainstream

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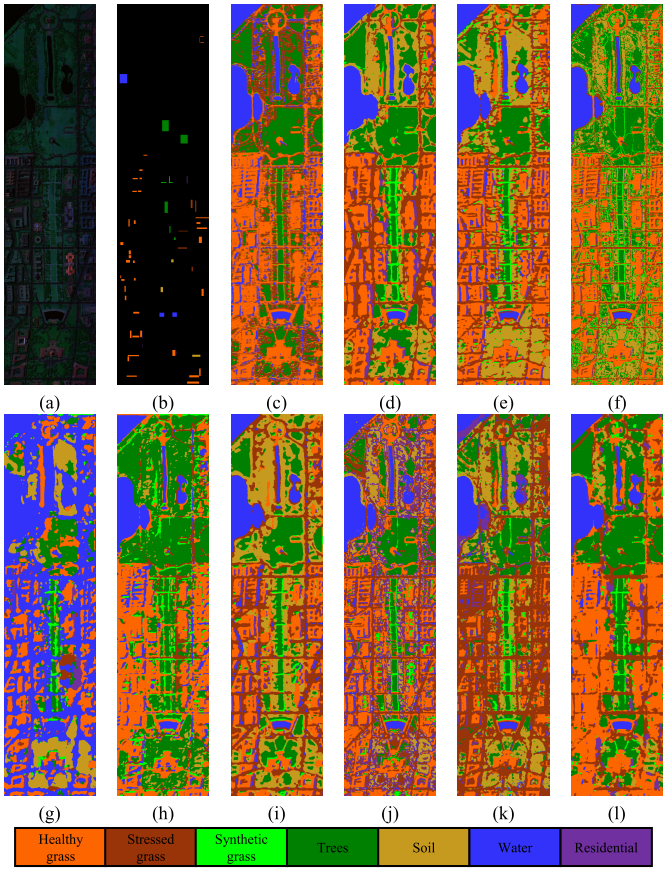


Fig. 9. Classification map on Washington DC MALL. (a) False-color Image. (b) GT. (c) UNet. (d) FreeNet. (e) FContNet. (f) SSFCN. (g) UperNet. (h) Segformer. (i) TransUNet. (j) DSNet. (k) SV2Net. (l) Ours.

models under conditions of limited training samples. Typically, the quantity of training data significantly influences model performance. To comprehensively examine model performance with scarce samples, we conducted experiments on four benchmark datasets, namely, Indian Pines, Houston 2013, Berlin, and Washington DC MALL, by setting distinct subsets of training samples. Specifically, for the IP dataset, the number of training samples was set to 1, 2, 5, 10, and 15. For the Houston 2013 dataset, we used 1, 2, 10, 15, and 20 training samples. The Berlin dataset was tested using 1, 25, 50, 100, and 200 training samples, while for the Washington DC MALL dataset, we chose subsets containing 1, 2, 5, 10, and 15 samples, respectively.

As shown in Fig. 10, the classification accuracy of MSUANet and the comparative methods across four datasets varies with the number of training samples. Overall, the accuracy of all models generally improves with an increasing number of training samples. However, certain methods, such as SSFCN, exhibit notably poor performance on the Indian Pines dataset, likely due to an inherent limitation in its learning capability. Specifically, on the relatively straightforward Washington DC MALL dataset, MSUANet is outperformed by FContNet and SSFCN when only one training sample per class is provided. This is likely attributable to the high spectral purity and pronounced interclass spectral differences in the Washington DC MALL dataset, enabling FContNet and SSFCN to converge more quickly and perform better with limited training data. In contrast, on datasets with substantial spectral mixing and overlap, such as Indian Pines, Houston 2013, and Berlin, MSUANet consistently achieves the best performance. This is mainly due to its use of multistage information sharing and adaptive latent unmixing information assistance, which help to capture the relationship between classification and unmixing tasks and enhance its ability to distinguish complex spectral signatures. Compared to methods directly fusing unmixing information, such as SV2Net and DSNet, MSUANet better utilizes the latent information gains arising from the unmixing task. As the number of training samples increases, this latent information becomes more abundant and refined, better adapting to the classification network and thus maximizing MSUANet's performance, consistently surpassing other methods across all datasets.

D. Parameter Sensitivity Analysis

To verify the stability of the uncertainty weighted loss, this section conducts a sensitivity analysis on the initial value of its key parameter σ^2. We set different initial values of $\sigma^2 \in [10^{-1}, 5^{-1}, 2^{-1}, 1, 2, 5, 10]$ and repeated the experiment in the same environment.

As shown in Fig. 11, we compared the classification results on four datasets under different σ^2 initializations, with the cosine-annealing strategy using equal task weights serving as the baseline. The experimental results show that different initial values of σ^2 have a minimal impact on the overall model performance. On the IP, HU, and DC datasets, the variations in OA, AA, and Kappa are all within $\Delta = 1.43$. Although the BE dataset suffers from severe spectral confusion, the variations of OA, AA, and Kappa remain within approximately $\Delta = 3.51$. Overall, the model exhibits good stability with respect to different initializations of σ^2. Compared with the baseline that adopts equal task weights and a cosine annealing schedule, the uncertainty-weighted approach achieves comparable or slightly better performance across all datasets. This demonstrates that jointly optimizing σ^2 through backpropagation effectively learns appropriate weighting coefficients, enabling the model to adaptively balance task uncertainties.

E. Visualization Analysis of MTL Structures

In this section, we perform a visual analysis to evaluate the effectiveness of the proposed MTL architecture for classification tasks. We design four representative task configurations (as shown in Fig. 12) for comparison: (a) single-task learning; (b) dual-task learning; (c) dual-task learning with simple feature fusion; and (d) our proposed MSUANet, which applies a multistage information sharing strategy. We use three types of visualization methods to examine the models' feature learning and class discrimination abilities: feature map visualization, t-distributed stochastic neighbor embedding (t-SNE) projection, and Grad-CAM class activation maps. These visual results help reveal how different task structures affect the learned representations and provide further insight into their influence on model behavior and classification outcomes.

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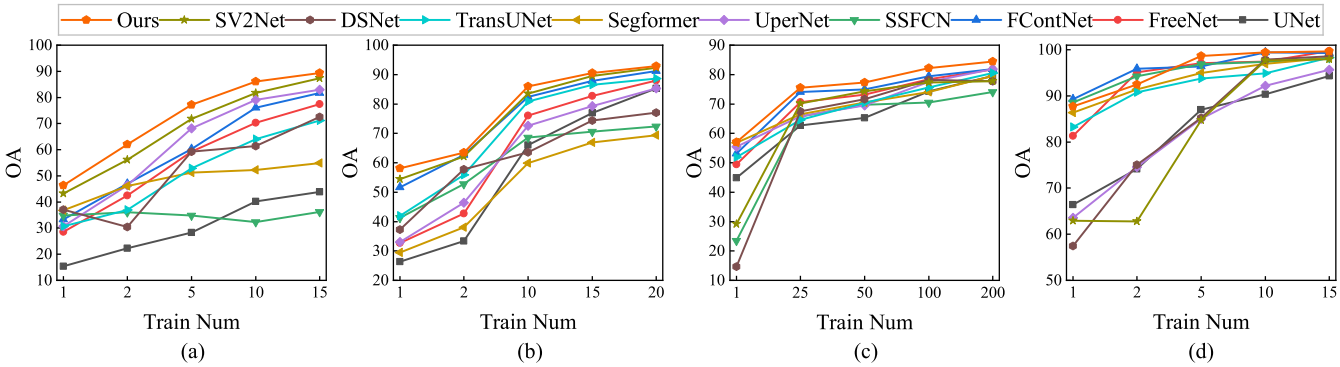


Fig. 10. Classification performance in terms of OA under different training sample sizes on four benchmark datasets. (a) Indian Pines, (b) Houston 2013, (c) Berlin, and (d) Washington DC MALL.

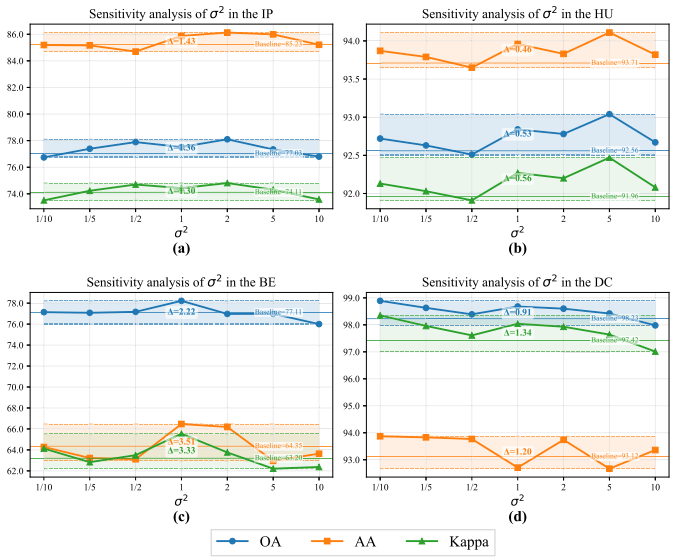


Fig. 11. Classification performance for different initial values of σ on the four datasets. (a) Indian Pines, (b) Houston 2013, (c) Berlin, and (d) Washington DC MALL.

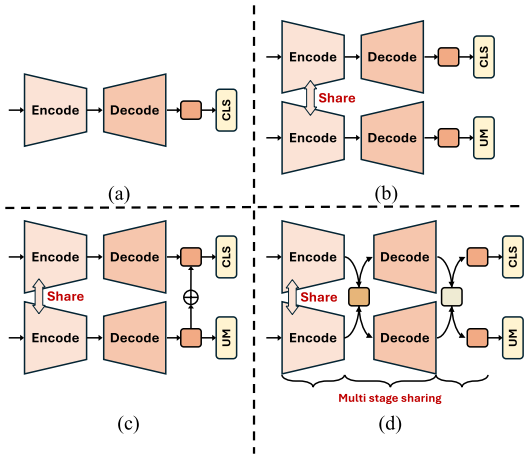


Fig. 12. Four representative task configurations for classification and unmixing. (a) Single-task network. (b) Dual-task network with independent decoding branches. (c) Dual-task network with direct fusion of unmixing information into the classification branch. (d) Proposed MSUANet incorporating multi-stage information sharing for enhanced task collaboration.

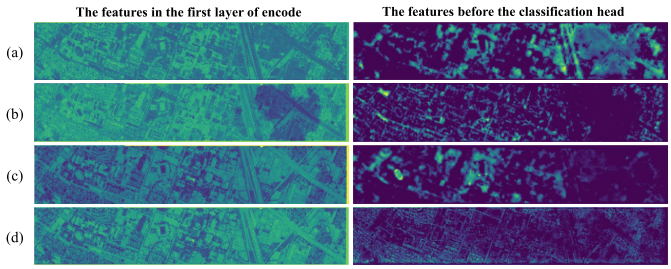


Fig. 13. Feature maps of the first encoder layer and the layer before the classification head on the Houston 2013 dataset. The left column shows the features from the first encoder layer, and the right column shows the features before the classification head. (a) Single-task network. (b) Dual-task network. (c) Dual-task network with direct fusion of unmixing information. (d) Proposed MSUANet.

1) Feature Map Visualization: To explore the feature extraction capabilities of different MTL structures for HSI, we visualize the feature maps from both shallow and deep network layers. To ensure fair comparison, we select the Houston 2013 dataset as input and extract the output from the first layer of the encoder and the last layer of the decoder across all architectures.

As shown in Fig. 13, the first layer outputs from the encoder demonstrate that all architectures can effectively extract structural features, with strong responses to roads and buildings. However, methods (a) and (b) exhibit less detailed textures compared to (c) and (d), likely due to the absence of unmixing mechanisms, which promote finer spectral representations. For the final decoder layer, method (a) shows a scattered focus on ground object details. Although method (b) slightly improves in precision, it remains relatively blurry. Method (c) exhibits better focus but weaker edge delineation. Our method (d), MSUANet, achieves superior boundary preservation and noise suppression, capturing target regions more reliably. In general, MSUANet demonstrates better structure preservation and semantic focus in both shallow and deep layers, validating its effectiveness under MTL architectures.

2) t-SNE Visualization: To gain deeper insight into the discriminative capabilities of different MTL structures, we apply t-SNE to the high-dimensional features from the decoder for 2-D visualization. By maintaining the local pairwise

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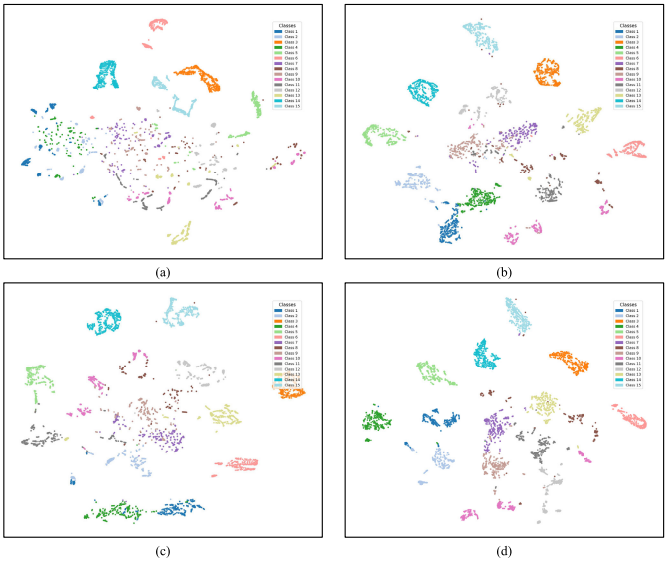


Fig. 14. t-SNE visualizations of class features extracted by different networks on the Houston 2013 dataset. (a) Single-task network. (b) Dual-task network. (c) Dual-task network with direct fusion and unmixing of information. (d) Proposed MSUANet.

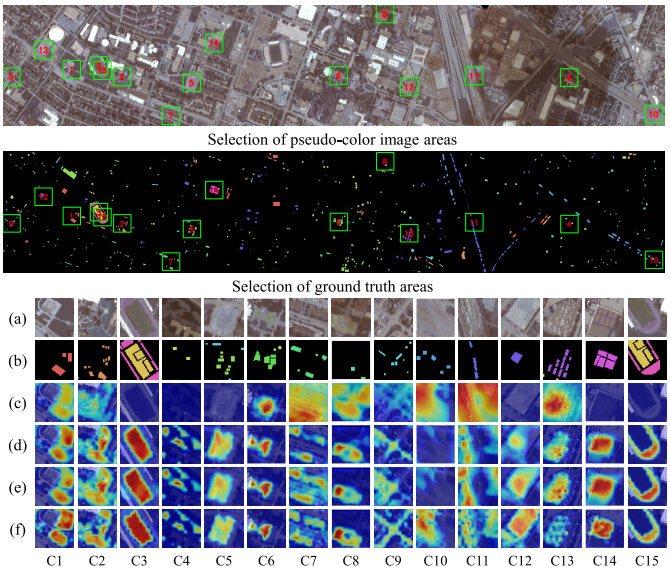


Fig. 15. Grad-CAM visualizations for different task structures. (Top) Selected false-color image and the corresponding ground-truth regions from the Houston 2013 dataset. (Bottom) Grad-CAM visualizations generated by different networks. (a) Selected false-color image. (b) Ground-truth regions. (c) Single-task network. (d) Dual-task network. (e) Dual-task network with direct fusion of unmixing information. (f) Proposed MSUANet.

similarities in the high-dimensional space, t-SNE embeds the features into a 2-D space, allowing the visualization of class-specific clustering and distribution trends.

As shown in Fig. 14, the single-task method (a) produces widely scattered features with severe interclass overlap and indistinct boundaries, indicating poor discriminability. Method (b) shows improved clustering, with some classes (Classes 3 and 14) forming compact groups, although significant overlap remains among others such as Classes 7, 9, and 11. Although method (c) integrates unmixing information directly into the network, its interclass feature separability is weaker than that of method (b), as indicated by more dispersed cluster centers and less well-defined class boundaries. This result suggests that the direct fusion of unmixing information does not inherently enhance the semantic discriminability of the learned features. In contrast, the MSUANet achieves the most favorable structure in terms of both intraclass compactness and interclass separability. The feature embeddings present a clustering form with a clear structure and definite boundaries, reflecting the model's stronger discriminative ability and focus on deep semantic feature learning.

3) Grad-CAM Visualization: To further investigate how the proposed MTL architecture influences the network's classification decisions, we employ Grad-CAM visualization to analyze class activation maps. Using the Houston 2013 dataset as an example, we select areas where classes can be easily distinguished, allowing us to visually illustrate differences among network models in terms of attention localization, feature discrimination, and semantic understanding capabilities.

As illustrated in Fig. 15, in categories 3, 4, 5, 12, 14, and 15, the single-task method (c) failed to effectively capture the crucial feature information necessary for accurate category discrimination, whereas the MTL methods (d)–(f) successfully highlighted the corresponding critical regions. Although the single-task method (a) demonstrated strong heatmap responses in categories 7, 8, and 11, a comparison against the ground-truth and pseudo color images reveals that this method tends to overly emphasize irrelevant background regions. In contrast, the MTL approaches exhibit more precise attention distributions, effectively concentrating on target areas. Moreover, taking the fine-grained category nine crossroads as an example, method (d) only partially attended to one of the roads, method (e) covered only certain portions of the intersection, and our proposed MSUANet comprehensively attended to the entire crossroad region. Experimental results indicate that MSUANet exhibits superior effectiveness in utilizing disentangled information. By incorporating the ALAKT module, the model can more effectively mine and leverage latent disentangled features, thereby enhancing the model's attention to target regions and its discriminative ability in HSI semantic feature extraction.

F. Explore the Concerns of Different Tasks on Information

In this section, we explore how the ALAKT-aided MSUANet adaptively selects the most relevant task-specific features, identifying critical information for both classification and unmixing tasks. We ran 100 times each on the Indian Pines, Houston 2013, Berlin, and Washington DC MALL datasets to record the average weights of the decoding features of each layer of ALAKT in the classification and unmixing tasks. The final weights and corresponding feature visualization results are shown in Fig. 16.

From Fig. 16, it can be observed that the weight distribution of the classification task is relatively uniform, and the distribution characteristics of the unmixing task are shifted to deep layers as a whole. For the Indian Pines, Houston 2013,

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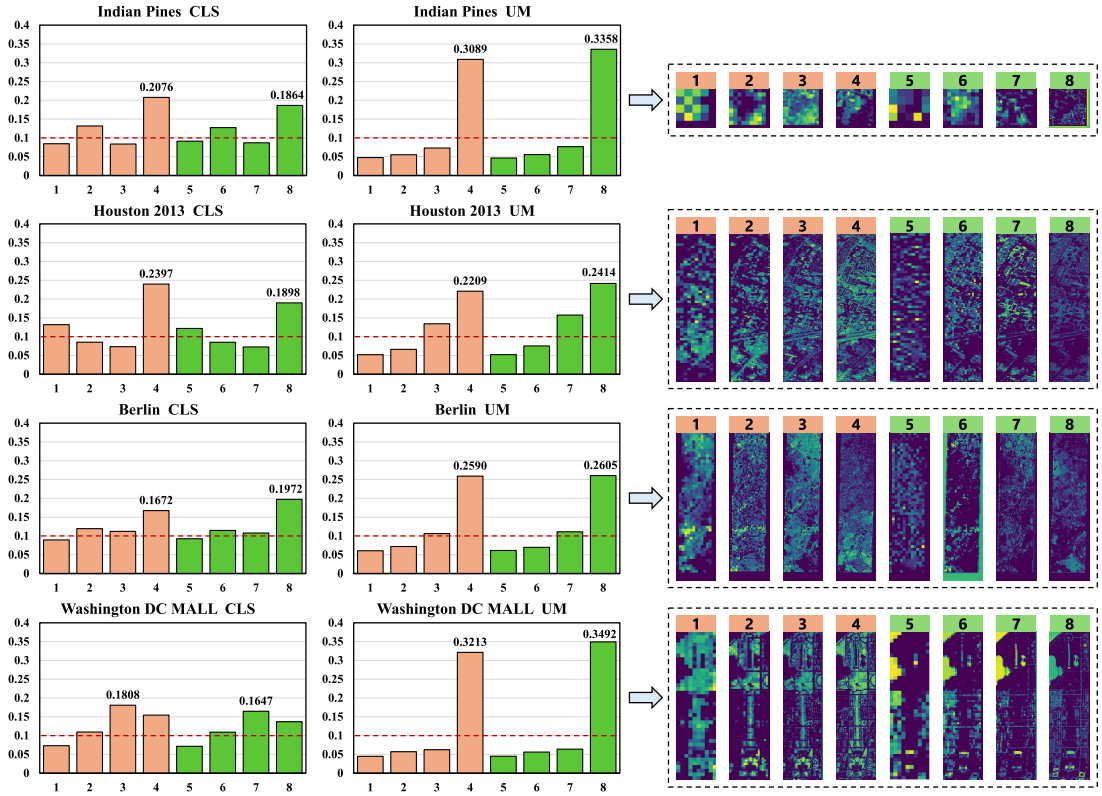


Fig. 16. Learned weight distributions and feature maps across decoding layers for classification and unmixing tasks. "CLS" is the classification task and "UM" is the unmixing task. Indices 1–4 correspond to classification decoding layers from shallow to deep, and indices 5–8 correspond to unmixing decoding layers from shallow to deep.

and Berlin datasets, the combined weights of the 4th and 8th layers exceed 36%, while the weights of other layers remain close to 10%. This indicates that the classification task mainly depends on deep layer features involved in classification and unmixing decode, while shallow and middle layer features play a complementary role. For the unmixing task, the proportion of deep layer weights exceeds 63% in the Indian Pines and Washington DC MALL datasets and reaches 46.23% and 51.95% for the Houston 2013 and Berlin, respectively. These results suggest that the unmixing task relies more heavily on deep layer feature representations. Feature map visualization further reveals that shallow layer features are relatively blurred, capturing only coarse outlines of land-cover information. As the decoder depth increases, the feature maps progressively delineate more precise object boundaries. Overall, the experimental results demonstrate that the classification task benefits from the synergistic integration of shallow, middle, and deep features, whereas the unmixing task depends more strongly on deep spectral semantic features.

G. Ablation Study

To validate the effectiveness of our proposed MSUANet, we designed and conducted a series of ablation experiments across four public datasets, with the experimental results summarized in Table IX. To thoroughly investigate the contributions of each module within MSUANet and evaluate the efficacy of the adopted multitask framework, we conducted five carefully designed comparative experiments: ① retaining only the classification branch in MSUANet, ② removing both the IDHA and ALAKT modules from the complete MSUANet, ③ removing only the ALAKT module from the complete MSUANet, ④ removing only the IDHA module from the complete MSUANet, and ⑤ employing the complete MSUANet.

TABLE IX
RESULTS OF ABLATION EXPERIMENTS ON FOUR DATASETS

Dataset	NO.	IDHA	ALAKT	MTL	OA	AA	Kappa
Indian Pines	①	✗	✗	✗	76.66	85.16	73.54
	②	✗	✗	✓	76.97	85.09	73.75
	③	✓	✗	✓	75.62	84.38	72.34
	④	✗	✓	✓	75.97	84.17	72.55
	⑤	✓	✓	✓	**77.52**	**85.57**	**74.43**
Houston 2013	①	✗	✗	✗	84.79	87.31	83.57
	②	✗	✗	✓	88.20	90.08	87.25
	③	✓	✗	✓	90.96	92.31	90.23
	④	✗	✓	✓	90.25	91.71	89.46
	⑤	✓	✓	✓	**92.84**	**93.96**	**92.27**
Berlin	①	✗	✗	✗	77.23	63.04	62.94
	②	✗	✗	✓	76.54	66.23	63.49
	③	✓	✗	✓	77.07	59.95	62.71
	④	✗	✓	✓	75.32	65.88	62.21
	⑤	✓	✓	✓	**78.22**	**66.47**	**65.54**
Washington DC MALL	①	✗	✗	✗	94.26	88.56	91.58
	②	✗	✗	✓	97.28	88.52	95.97
	③	✓	✗	✓	97.84	92.41	96.81
	④	✗	✓	✓	98.31	92.00	97.50
	⑤	✓	✓	✓	**98.68**	**92.71**	**98.04**

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TABLE X
COMPUTATIONAL COMPLEXITY AND DEPLOYABILITY OF EACH MODEL

Model	Param (M)	GFLOPs (G)	CPU Latency (ms)	GPU Latency (ms)	Peak Memory (MB)	Throughput (sample/s)
UNet	17.38	15.16	25.28	1.54	121.90	639.02
FreeNet	2.35	4.28	9.17	1.38	55.69	717.01
FContNet	1.25	8.44	49.66	22.69	254.81	43.91
SSFCN	0.11	2.28	12.15	0.37	99.23	2367.14
UperNet	28.04	7.64	28.14	3.09	214.95	323.82
Segformer	8.02	6.91	99.74	3.53	360.82	283.61
TransUNet	93.23	12.64	63.92	7.12	468.64	141.22
DSNet	200.47	6.93	52.03	1.56	855.88	641.89
SV2Net	1.55	4.13	182.97	4.17	227.70	284.15
MSUANet(w/o IDHA)	30.17	5.65	44.39	7.10	256.83	141.19
MSUANet(w/o ALAKT)	31.47	7.11	49.48	6.76	596.76	148.16
MSUANet	31.36	6.49	49.53	8.68	262.88	115.53

As shown in Table IX, we observed that on all four datasets, the MTL framework without IDHA and ALAKT Group ② consistently achieved higher OA, AA, and Kappa scores compared to the single-task model Group ①, highlighting the clear benefits of employing an MTL approach for HSI classification. The experimental results on the Indian Pines dataset indicate that individually introducing either IDHA or ALAKT modules does not significantly improve performance. However, when both modules are combined, optimal performance is achieved. This finding demonstrates the complementary nature between the IDHA and ALAKT modules, highlighting their collaborative ability to jointly enhance model performance. On datasets such as Houston 2013, Berlin, and Washington DC MALL, the inclusion of the IDHA module consistently outperforms both single-task and basic MTL approaches, signifying IDHA's effectiveness in capturing physically meaningful spectral components from large-scale HSI datasets. Additionally, we observed relatively poorer performance with the standalone ALAKT module on the Berlin dataset, potentially due to noisy or inaccurate labels hindering the accurate extraction of higher level semantics and latent abundance features, thus impairing adaptive feature integration. Overall, the proposed MSUANet leverages the combined strengths of IDHA for task-specific feature extraction and ALAKT for adaptive cross-task feature integration within an MTL framework, thereby significantly enhancing the HSI classification capability.

H. Analysis of Computational Complexity and Deployability

In this section, we analyze the computational complexity and deployability of the proposed MSUANet compared with mainstream methods. Table X summarizes the results in terms of the number of parameters, computational complexity, wall-clock latency on both CPU and GPU, peak memory usage, and throughput. All measurements were conducted on an Intel i9-13900K CPU and an RTX 4090 D GPU under identical input tensor [1, 200, 145, 145].

Compared to classical encoder-to-decoder architectures such as UNet and UperNet, MSUANet achieves a favorable tradeoff between accuracy and efficiency. Although MSUANet contains a slightly higher parameter count 31.36 M, its GFLOPs, 6.49 G, remain moderate, lower than UperNet 7.64 G and comparable to Segformer 6.91 G. The GPU inference latency of 8.68 ms corresponds to roughly 115 FPS, which satisfies real-time processing requirements. On CPU, the latency of 49.53 ms indicates a near real-time capability for HSI. In terms of memory and throughput, MSUANet requires 262.88 MB of peak GPU memory and processes 115.53 sample/s, achieving satisfactory efficiency. Module-level ablation indicates that removing IDHA leads to an increase of 1.12 M parameters and 0.84 GFLOPs, while removing ALAKT results in an increase of 0.11M parameters and 0.62 GFLOPs. This observation may be attributed to the attention mechanism in IDHA, which introduces a small number of additional parameters, whereas ALAKT helps MSUANet filter redundant features, thereby reducing computational cost. These results demonstrate that MSUANet is deployable in real-time and edge scenarios. On a desktop GPU, it comfortably exceeds 100 FPS, and even on resource-constrained devices, further optimization via mixed-precision inference and pruning can enable practical real-time deployment.

I. Analysis of the Effectiveness of IDHA

To evaluate the effectiveness of each component within the IDHA, we conducted item-by-item ablation experiments under MSUANet with only IDHA. Specifically, the experiment successively removed or replaced different components in the module, including 3-D position encoding, attention gating, activation functions, and residual connections, to analyze their impact on overall performance. As shown in Table XI, Groups 1–6 correspond to the following operations respectively—Group 1: remove the 3-D position encoding, Group 2: replace the tanh activation function with Softmax, Group: 3 replace the tanh activation function with dot product, Group 4: remove the residual connections, Group 5: retain only spectral attention, and Group 6: retain only spatial attention. Here, Δ denotes the performance difference from the baseline, while Δm represents the averaged performance gap across the four datasets.

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TABLE XI
OA RESULTS OF THE SINGLE ABLATION EXPERIMENT FOR IDHA ON THE IP, HU, BE, AND DC DATASETS

	IP	Δ	HU	Δ	BE	Δ	DC	Δ	Δm
base	75.62	0	90.96	0	77.07	0	97.84	0	0
1	75.19	0.43	90.32	0.64	76.25	0.82	97.23	0.61	0.62
2	75.54	0.08	90.30	0.66	76.41	0.66	97.66	0.18	0.39
3	75.46	0.16	90.19	0.77	76.15	0.92	97.34	0.50	0.58
4	51.04	24.58	59.06	31.9	70.05	7.02	82.79	15.05	19.63
5	73.58	2.04	90.18	0.78	76.01	1.06	97.57	0.27	1.03
6	74.65	0.97	90.39	0.57	76.25	0.82	97.63	0.21	0.64

The experimental results indicate that the relative importance of each component follows the order: 4 residual connection > 5 spectral attention > 6 spatial attention > 1 3-D positional encoding > 2 softmanx > 3 dot product. Removing the residual connection leads to the most significant performance degradation, with accuracy drops of 31.9 on HU and 24.58 on IP, suggesting that low-level feature encoding plays a crucial role in subsequent feature fusion. Retaining only spectral or spatial attention also results in notable performance deterioration, demonstrating their complementary nature in information representation. The removal of 3-D positional encoding causes only a slight decline, indicating that spatial spectral positional priors contribute effectively to HSI feature modeling. Replacing tanh with Softmax or dot product slightly decreases performance, implying that the tanh function enables smoother feature distributions. Overall, the improvement achieved by the IDHA does not stem from any single structural design but rather from the synergistic interaction among hybrid attention mechanisms, positional encoding, and residual connections.

J. Analysis of the Effectiveness of ALAKT

To verify the effectiveness of the proposed ALAKT, we conducted a series of ablation experiments to analyze the impact of different fusion weighting strategies and branch configurations on the overall model performance. Specifically, the following configurations were compared: using only the classification branch (CLS), adding the unmixing branch (CLS+UM), applying uniform-weight fusion (Fuse-Uniform), applying adaptive learnable-weight fusion (Fuse-ALAKT), incorporating the IDHA with uniform-weight fusion (IDHA-Uniform), and incorporating the IDHA with adaptive learnable-weight fusion (IDHA-ALAKT). The experimental results are illustrated in Fig. 17.

The experimental results demonstrate that incorporating the unmixing branch into the classification framework significantly improves performance across multiple datasets, with notable gains on Houston 2013 +3.41% and Washington DC MALL +3.02%. This indicates that the unmixing branch provides beneficial spectral and spatial complementary information to the classification branch. Compared with the uniform weights, the adaptive fusion introducing ALAKT has brought further performance improvements on all datasets. This indicates that the learnable adaptive weights can dynamically allocate the fusion weights based on the importance of features at different layers, thereby obtaining better feature representations. We further observed that after integrating the IDHA, both uniform and adaptive fusion achieved additional performance gains, with the adaptive fusion showing a more pronounced improvement. This finding indicates that IDHA and ALAKT complement each other in feature extraction. However, for the Indian Pines and Berlin datasets, both uniform and adaptive fusion without IDHA performed slightly worse than the CLS+UM configuration, likely due to redundancy and interference between classification and unmixing information. The inclusion of IDHA effectively mitigates such cross-task interference. In summary, the performance improvements achieved by ALAKT are not attributable to a single design component, but rather to the synergistic effect of multilevel adaptive feature fusion and the complementary interaction with the IDHA.

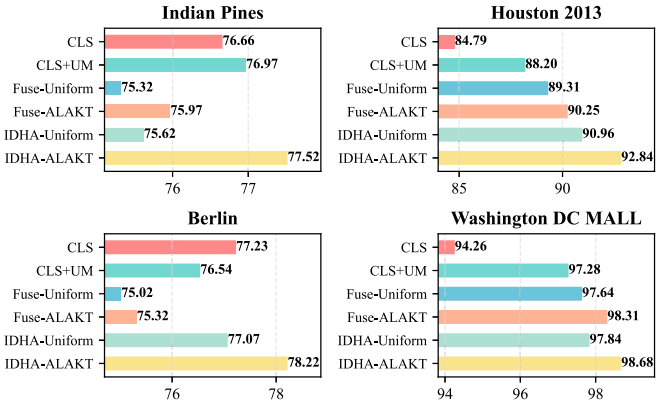


Fig. 17. OA results of the single ablation experiment for ALAKT on the IP, HU, BE, and DC datasets.

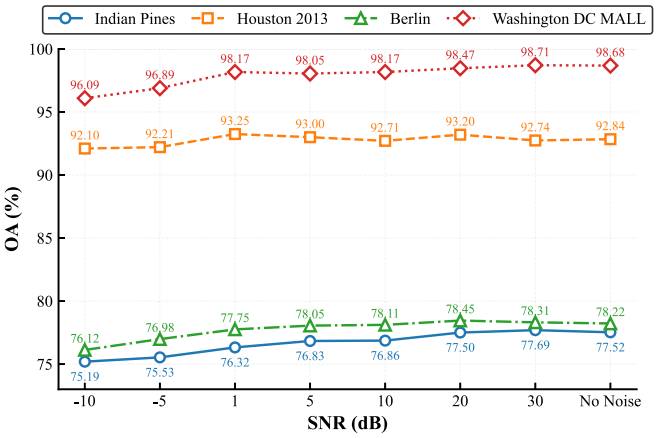


Fig. 18. OA results of different noise abundance maps of ALAKT on IP, HU, BE, and DC.

K. Reliability Analysis of Unmixed Labels

To quantitatively assess the reliability and robustness of these pseudo labels, Gaussian noise was added to the abundance maps with different signal-to-noise ratios (SNRs) ranging from −10 to 30 dB, as well as a noiseless reference case. The OA variations across different SNRs are shown in Fig. 18, for the Indian Pines, Houston 2013, Berlin, and Washington DC MALL datasets.

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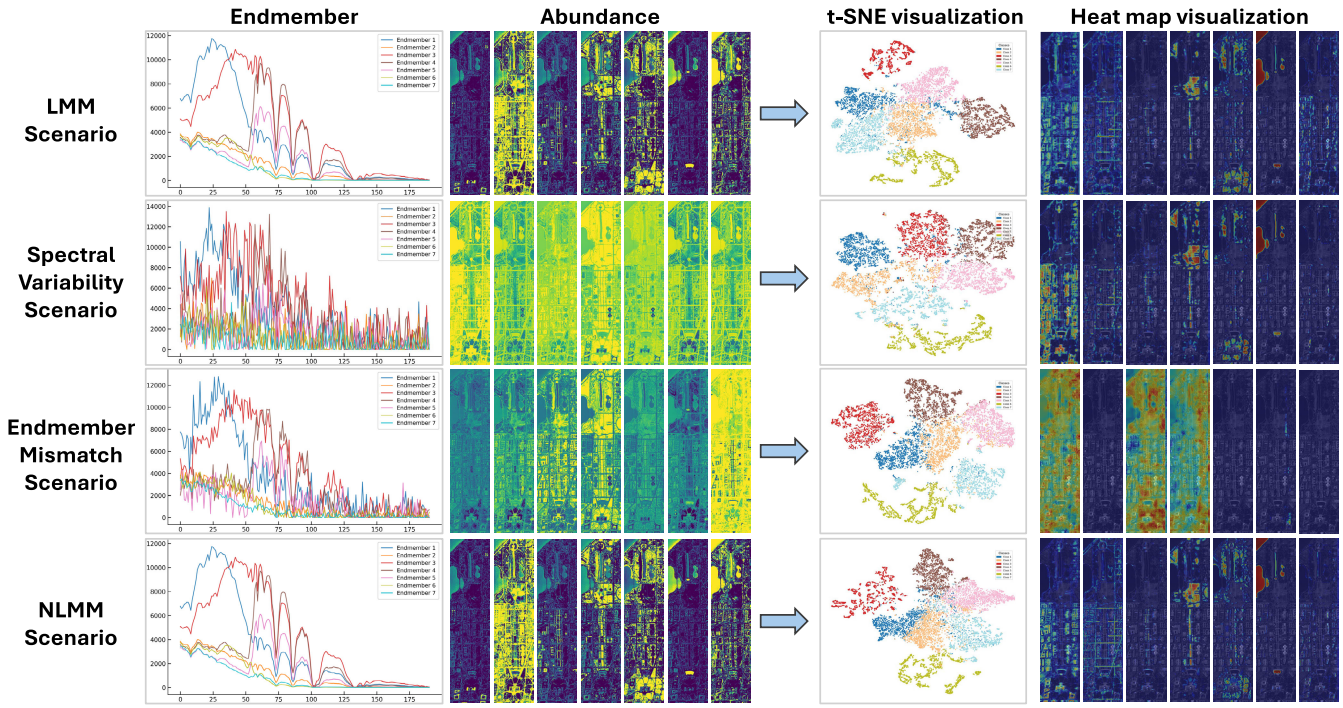


Fig. 19. Comparative visualization of MSUANet outputs under LMM, spectral variability, endmember mismatch, and NLMM scenarios.

The results indicate that as the SNR increases, the OA of all datasets gradually becomes more stable. When the power of the abundance signal is comparable to that of the noise (SNR = 1 dB), the model shows a decrease of −1.20% on Indian Pines, an increase of +0.41% on Houston 2013, a decrease of −0.47% on Berlin, and a decrease of −0.51% on the Washington DC MALL. This demonstrates that moderate perturbations in pseudo labels have only a limited impact on the model's performance. When the abundance signal power is only one-tenth of the noise power (SNR = −10 dB), the model experiences accuracy drops of −2.33%, −0.74%, −2.10%, and −2.59% on the four datasets compared with the noiseless condition. These results reveal that the model still maintains strong robustness even under severe label noise. Overall, the abundance labels generated from spectral unmixing are not precise annotations but rather coarse soft labels. The coarseness of such labels naturally provides a degree of robustness to noise. Instead of becoming a performance bottleneck, this form of coarse supervision enhances the model's stability and generalization capability in complex and realistic HSI scenarios.

L. Limitations and Robustness

1) Behavior Under Complex Spectral Conditions: While MSUANet demonstrates strong performance on real HSI datasets, these datasets typically contain a limited degree of spectral distortion and do not fully cover the extreme spectral behaviors that may appear in practical applications. To comprehensively evaluate the robustness of the model in complex spectral situations, we construct three types of scenarios: severe spectral variation scenario, endmember mismatch scenario, and NLMM scenario. Fig. 19 illustrates the endmembers and abundances under the three constructed spectral scenarios on the DC dataset, along with the t-SNE feature embeddings and heat map responses produced by the trained MSUANet.

The experimental results show that in the LMM scenario, the endmember spectra exhibit high separability, and the abundance maps exhibit clear region boundaries. Correspondingly, the t-SNE features form compact and well-separated clusters, and the heat maps focus primarily on the correct classes. This indicates that MSUANet can effectively utilize unmixing information for reliable discrimination. In the spectral variation and endmember mismatch scenarios, although the endmember signatures and abundance distributions undergo varying degrees of distortion and irregularity, the corresponding t-SNE feature distributions still maintain good interclass separability. From the heat maps, we observe that endmember mismatch leads to a more noticeable attention drift compared with the spectral variation case. This indicates that spectral variations and endmember deviations introduce some disturbances to the model, while MSUANet still maintains a relatively stable feature structure and discriminative capability. Notably, under the NLMM scenario, the feature distributions extracted by MSUANet do not exhibit drastic changes compared with the linear case. Only slight cluster expansion or mild broadening of attention regions can be observed for a few classes. This demonstrates that, benefiting from the adaptive task selection mechanism of ALAKT, MSUANet is still capable of extracting discriminative features under nonlinear mixing conditions, showing strong robustness against a certain degree of nonlinearity.

2) NLMM Stress Test: To further evaluate the robustness of MSUANet under nonlinear mixing conditions, we

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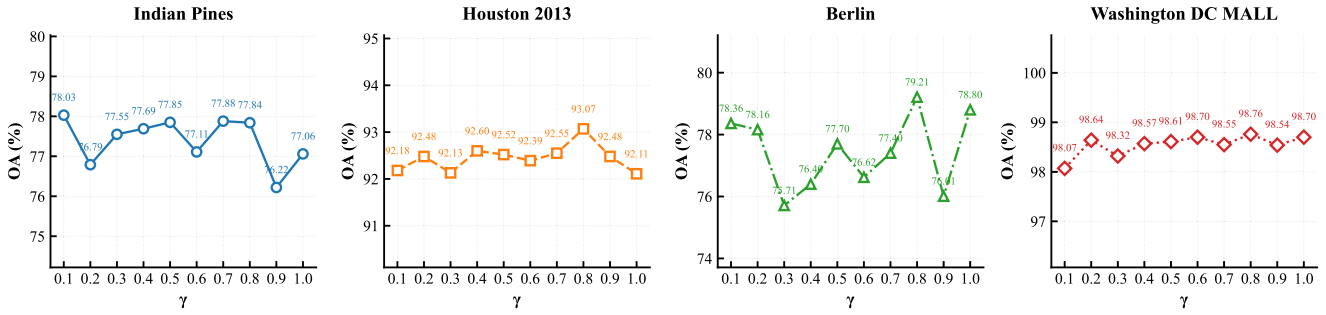


Fig. 20. OA of MSUANet under different NLMM interaction strengths γ on the IP, HU, BE, and DC.

construct NLMM datasets based on the generalized bilinear model for all benchmark scenes, as shown in (25). Specifically, the interaction coefficient γ is gradually increased from 0.1 to 1.0 to generate a series of nonlinear mixtures ranging from weak to strong nonlinearity, as illustrated in Fig. 20

$$\mathbf{Y} = \mathbf{M}\mathbf{a} + \sum_{i<j} \gamma a_i a_j \left(\mathbf{m}_i \odot \mathbf{m}_j\right) \tag{25}$$

where γ controls the interaction strength between materials, $a_i a_j(\mathbf{m}_i \odot \mathbf{m}_j)$ represents the bilinear interaction between endmembers i and j, $a_i a_j$ denotes the joint abundance contribution of the two materials, and $\mathbf{m}_i \odot \mathbf{m}_j$ denotes the Hadamard product of their spectral signatures.

The experimental results show that as the degree of nonlinearity increases, the OA exhibits only minor fluctuations across all datasets, without a significant performance drop. The IP and BE datasets display slight variations but remain generally stable, while the HU and DC datasets are almost unaffected across all levels of nonlinearity. This trend is consistent with the t-SNE and heat map visualizations in Fig. 19, further demonstrating the robustness of MSUANet under nonlinear mixing conditions.

V. Conclusion

In this article, we propose an MSUANet for HSI classification based on MTL, aiming to alleviate task interference during MTL information sharing and enhance the efficient utilization of task-relevant information. Specifically, MSUANet leverages a multistage sharing mechanism to fully exploit the complementary information across different tasks, thereby enhancing the classification performance of the primary task. In addition, we design IDHA and ALAKT to further strengthen the representation of task-specific information and to adaptively select complementary information. Extensive experiments on four benchmark datasets demonstrate the superior performance of MSUANet compared to state-of-the-art methods.

In the future, we plan to further explore model compression, knowledge distillation, and binary neural networks to develop more lightweight versions of our model, thereby improving its suitability for deployment on resource-constrained platforms.

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